

so that the $SL_n(\mathbb{R})$ -action on X_0 is proper, $SO_n(\mathbb{R})$ being compact. Moreover, X_0 is a contractible manifold; in fact, the interested reader can verify that X_0 is diffeomorphic to Euclidean space of dimension $n(n+1)/2 - 1$. As in example 6, then, we have $H_*G \approx H_*(X_0/G)$ for any discrete, torsion-free subgroup $G \subset SL_n(\mathbb{R})$. We note that if $n = 2$ the space X_0 can be identified with the upper half plane $\mathcal{H} \subset \mathbb{C}$, with $SL_2(\mathbb{R})$ acting by linear fractional transformations:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} z = \frac{az + b}{cz + d}.$$

Indeed, one checks that this defines a transitive action of $SL_2(\mathbb{R})$ on $\mathcal{H}$ and that the isotropy group at $z = i$ is $SO_2(\mathbb{R})$; this leads easily to the desired homeomorphism

$$\mathcal{H} \approx SL_2(\mathbb{R})/SO_2(\mathbb{R}) \approx X_0.$$

8. Finally, we state the general facts which were illustrated in the previous examples. Let L be a Lie group with only finitely many connected components. Then L has a maximal compact subgroup K (unique up to conjugacy), and the homogeneous space $X = L/K$ is diffeomorphic to $\mathbb{R}^d$, $d = \dim L - \dim K$. (Proofs can be found, for example, in Hochschild [1965], ch. XV.) Consequently, if $G \subset L$ is a discrete, torsion-free subgroup, then the quotient manifold X/G is a $K(G, 1)$ and $H_*G \approx H_*(X/G)$.

EXERCISES

1. Let Y be a path-connected space. If Y has a contractible, regular covering space X with covering group G , show that $H_*Y \approx H_*G$. [Hint: The singular chain complex $C_*^{\text{sing}}(X)$ provides a free resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ and $C_*^{\text{sing}}(X)_G \approx C_*^{\text{sing}}(Y)$.]
2. (This exercise requires some elementary homotopy theory.) Let Y be a $K(G, 1)$ -space, i.e., a path-connected space with $\pi_1 Y = G$ and $\pi_i Y = 0$ for $i > 1$. Prove that $H_*Y \approx H_*G$. [Hint: This is clear if Y is a CW-complex; in the general case, replace Y by a CW-complex which is weakly homotopy equivalent to Y , cf. Spanier [1966], §7.8. Alternatively, one can directly construct a chain homotopy equivalence $C_*^{\text{sing}}(Y) \simeq C_*G$, cf. Eilenberg–MacLane [1945].] Note that exercise 1 is a special case of the present exercise.
3. Fix an integer $n \geq 1$. For any integer $N \geq 2$, let $\Gamma(N)$ be the kernel of the canonical map $GL_n(\mathbb{Z}) \rightarrow GL_n(\mathbb{Z}/N\mathbb{Z})$, i.e., $\Gamma(N) = \{g \in GL_n(\mathbb{Z}) : g \equiv 1 \pmod{N}\}$, where 1 denotes the identity matrix. The group $\Gamma(N)$ is called the *principal congruence subgroup of $GL_n(\mathbb{Z})$ of level N* . Note that $\Gamma(N)$ has finite index in $GL_n(\mathbb{Z})$, since $GL_n(\mathbb{Z}/N\mathbb{Z})$ is finite. The purpose of this exercise is to prove that $\Gamma(N)$ is torsion-free for $N \geq 3$.
 - (a) Let p be a fixed prime and let A be an $n \times n$ matrix of integers such that $A \equiv 1 \pmod{p}$. If $A \neq 1$, then there is a unique positive integer $d = d(A)$ such that

$$A \equiv 1 \pmod{p^d} \quad \text{and} \quad A \not\equiv 1 \pmod{p^{d+1}}.$$

Show that $d(A^q) = d(A)$ for any prime $q \neq p$. If p is odd or $d(A) \geq 2$, show that $d(A^p) = d(A) + 1$. [Hint: Write $A = 1 + p^d B$ with $B \not\equiv 0 \pmod{p}$, and look at the binomial expansion of $(1 + p^d B)^l$, $l = p$ or q .]

(b) Deduce that $\Gamma(N)$ is torsion-free for $N \geq 3$ and that $\Gamma(2)$ has only 2-torsion. [Hint: Suppose $A \in \Gamma(N)$ has prime order, and apply (a) with p a prime divisor of N .]

5 Hopf's Theorems

The purpose of this section is to prove the results stated as 0.1 and 0.2 in the introduction. We will need to use the Hurewicz theorem (cf. Spanier [1966], ch. 7, §5), which says that if $\pi_i X = 0$ for $i < n$ (where $n \geq 2$), then $H_i X = 0$ for $0 < i < n$ and the Hurewicz map $h: \pi_n X \rightarrow H_n X$ is an isomorphism. (In fact, an examination of our proofs will show that we only need the surjectivity of h , which is considerably easier to prove; indeed, it follows directly from Spanier's Thm. 7.4.8.)

We begin by observing that for any group G and integer n one can compute $H_i G$ for $i \leq n + 1$ from a *partial* projective resolution of length n :

(5.1) Lemma. *Let $F_n \rightarrow \cdots \rightarrow F_0 \rightarrow \mathbb{Z} \rightarrow 0$ be an exact sequence of $\mathbb{Z}G$ -modules where each F_i is projective. Then $H_i G \approx H_i(F_G)$ for $i < n$ and there is an exact sequence*

$$0 \rightarrow H_{n+1}(G) \rightarrow (H_n F)_G \rightarrow H_n(F_G) \rightarrow H_n(G) \rightarrow 0.$$

PROOF. Extend F to a full resolution F^+ by choosing a projective module F'_{n+1} mapping onto $H_n F$, etc.:

$$\begin{array}{ccccccccccc} \cdots & \rightarrow & F'_{n+2} & \rightarrow & F'_{n+1} & \rightarrow & F_n & \rightarrow & F_{n-1} & \rightarrow & \cdots & \rightarrow & F_0 & \rightarrow & \mathbb{Z} & \rightarrow & 0. \\ & & & & \searrow & & \uparrow & & & & & & & & & & \\ & & & & & & H_n F & & & & & & & & & & \end{array}$$

It is easy to see (either by direct inspection or by considering the homology exact sequence associated to the exact sequence $0 \rightarrow F_G \rightarrow F_G^+ \rightarrow F_G^+/F_G \rightarrow 0$ of chain complexes) that $H_i(F_G^+) = H_i(F_G)$ for $i < n$ and that there is an exact sequence

$$0 \rightarrow H_{n+1}(F_G^+) \rightarrow A \rightarrow H_n(F_G) \rightarrow H_n(F_G^+) \rightarrow 0,$$

where $A = \text{coker} \{(F'_{n+2})_G \rightarrow (F'_{n+1})_G\}$. By 2.2 this cokernel can be identified with $(H_n F)_G$, whence the lemma. $\square$

(5.2) Theorem. *For any connected CW-complex Y there is a canonical map $\psi: H_* Y \rightarrow H_* \pi$ ($\pi = \pi_1 Y$). If $\pi_i Y = 0$ for $1 < i < n$ (for some $n \geq 2$) then ψ is an isomorphism $H_i Y \xrightarrow{\sim} H_i \pi$ for $i < n$, and the sequence*

$$\pi_n Y \xrightarrow{h} H_n Y \xrightarrow{\psi} H_n \pi \rightarrow 0$$

is exact.

(Note that the hypothesis always holds with $n = 2$, so 5.2 yields the result 0.1 stated in the introduction.)

PROOF. Let X be the universal cover of Y and let F be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\pi$. Since $C_*(X)$ is a complex of free $\mathbb{Z}\pi$ -modules augmented over $\mathbb{Z}$, the fundamental lemma (I.7.4) gives us a chain map (over $\mathbb{Z}\pi$) $C_*(X) \rightarrow F$, well-defined up to homotopy. Taking co-invariants we obtain a map $C_*(Y) \rightarrow F_\pi$, which induces the desired map $\psi: H_* Y \rightarrow H_* \pi$. One knows that $\pi_i X \cong \pi_i Y$ for $i > 1$ (cf. Spanier [1966], 7.3.7), so our hypothesis implies that $\pi_i X = 0$ for $i < n$. Hence $H_i X = 0$ for $0 < i < n$ and the Hurewicz map $h: \pi_n X \rightarrow H_n X$ is an isomorphism. We therefore have a partial free resolution

$$C_n(X) \rightarrow \cdots \rightarrow C_0(X) \rightarrow \mathbb{Z} \rightarrow 0,$$

whose n -th homology group is the group $Z_n X$ of n -cycles of X . Lemma 5.1 now implies that $H_i Y \cong H_i \pi$ for $i < n$ and that there is an exact sequence

$$Z_n X \rightarrow Z_n Y \xrightarrow{\tilde{\psi}} H_n \pi \rightarrow 0.$$

The map $\tilde{\psi}$ which arises here is easily seen to be the composite $Z_n Y \rightarrow H_n Y \xrightarrow{\psi} H_n \pi$. Consequently the sequence

$$H_n X \rightarrow H_n Y \xrightarrow{\psi} H_n \pi \rightarrow 0$$

is exact, and the desired exact sequence

$$\pi_n Y \xrightarrow{h} H_n Y \xrightarrow{\psi} H_n \pi \rightarrow 0$$

now follows from the diagram

$$\begin{array}{ccc} \pi_n X & \xrightarrow{\quad h \quad} & H_n X \\ \downarrow \approx & & \downarrow \\ \pi_n Y & \xrightarrow{\quad h \quad} & H_n Y. \end{array}$$

□

We turn now to formula 0.2 of the introduction.

(5.3) Theorem. *If $G = F/R$ where F is free, then $H_2 G \approx R \cap [F, F]/[F, R]$.*

PROOF. Let $F = F(S)$, let Y be a bouquet of circles indexed by S , and let $\tilde{Y}$ be the connected regular covering space of Y corresponding to the normal subgroup R of $F(S) = \pi_1 Y$. Choosing a basepoint $\tilde{v}$ in $\tilde{Y}$ lying over the vertex of Y , we identify $G = F/R$ with the group of covering transformations of $\tilde{Y}$ as in Chapter I, Appendix, A1. For any $f \in F$ we regard f as a combinatorial path in Y and we denote by $\tilde{f}$ the lifting of f to $\tilde{Y}$ starting at $\tilde{v}$. [By a *combinatorial path* in a CW-complex we mean a sequence $e_1, \dots, e_n$ of oriented 1-cells such that the initial vertex of e_{i+1} is equal to the final vertex of e_i for $i = 1, \dots, n - 1$.] This path $\tilde{f}$, then, ends at the vertex $\tilde{f}\tilde{v}$, where $\tilde{f}$ is the image of f in G .

The complex $C_* \tilde{Y}$ is a complex of free $\mathbb{Z}G$ -modules, and it provides a partial resolution $C_1 \tilde{Y} \rightarrow C_0 \tilde{Y} \rightarrow \mathbb{Z} \rightarrow 0$. We can therefore apply 5.1 to obtain $H_2 G \approx \ker\{(H_1 \tilde{Y})_G \rightarrow H_1 Y\}$. Now $H_1 \tilde{Y} \approx (\pi_1 \tilde{Y})_{ab} \approx R_{ab}$, and I claim that the composite isomorphism $H_1 \tilde{Y} \approx R_{ab}$ is an isomorphism of G -modules, where the G -action on R_{ab} is induced by the conjugation action of F on R . [This makes sense because the conjugation action of R on itself induces the trivial action of R on R_{ab} .] To verify the claim, one checks the definitions and finds that the isomorphism $R_{ab} \xrightarrow{\sim} H_1 \tilde{Y}$ is induced by a map $d: R \rightarrow H_1 \tilde{Y}$ defined as follows: For any $r \in R$ the lifting $\tilde{r}$ is a closed path in $\tilde{Y}$; taking the sum of the oriented 1-cells which occur in $\tilde{r}$, we obtain a 1-cycle in $\tilde{Y}$, hence an element of $H_1 \tilde{Y}$, and this element is by definition dr . Now if $f \in F$ and $r \in R$, the lifting of frf^{-1} is the path

$$\begin{array}{ccccccc} & & \tilde{f} & & \tilde{f}\tilde{r} & & \tilde{f}^{-1} \\ & \circ & \text{---} & \circ & \text{---} & \circ & \text{---} & \circ \\ \tilde{v} & & \tilde{f}\tilde{v} & & \tilde{f}\tilde{v} & & \tilde{v} \end{array}$$

hence $d(frf^{-1}) = \tilde{f}dr$. This shows that the isomorphism $R_{ab} \xrightarrow{\sim} H_1 \tilde{Y}$ is an isomorphism of G -modules, as claimed.

We therefore have a diagram

$$\begin{array}{ccccc} (H_1 \tilde{Y})_G & \approx & (R_{ab})_G & = & R/[F, R] \\ \downarrow & & \downarrow & & \downarrow \\ H_1 Y & \approx & F_{ab} & = & F/[F, F] \end{array}$$

where the second and third vertical arrows are induced by the inclusion $R \hookrightarrow F$. Hence $H_2 G \approx \ker\{R/[F, R] \rightarrow F/[F, F]\} = R \cap [F, F]/[F, R]$. $\square$

(The G -module R_{ab} which occurred in the above proof is called the *relation module* associated to the presentation of G as F/R .)

Note that the chain complex $C_* \tilde{Y}$ can be described explicitly, exactly as in I.4.3. We therefore obtain, as a corollary of the above proof, the following result:

(5.4) Proposition. *If $G = F(S)/R$ then there is an exact sequence*

$$0 \rightarrow R_{ab} \xrightarrow{\theta} \mathbb{Z}G^{(S)} \xrightarrow{\partial} \mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

of G -modules, where $\mathbb{Z}G^{(S)}$ is free with basis $(e_s)_{s \in S}$, and $\partial e_s = \bar{s} - 1$.

(Here $\bar{s}$, as usual, denotes the image of s in G .)

The map θ which occurs here has been described explicitly in the proof of 5.3 in terms of path lifting. We will see in exercise 3d below that θ can also be described purely algebraically, in terms of the "free differential calculus."

Remarks

1. It is possible to give an algebraic proof of 5.4 (see exercise 4d of §IV.2) and then deduce Hopf's formula 5.3 directly from 5.4.

2. Hopf's formula suggests that, roughly speaking, $H_2 G$ consists of commutator relations $\prod [a_i, b_i] = 1$ in G , modulo those relations that hold trivially. See C. Miller [1952] for a precise formulation and proof of this statement.

EXERCISES

1. In the situation of 5.2, suppose in addition that Y is n -dimensional; prove that there is an exact sequence

$$0 \rightarrow H_{n+1} \pi \rightarrow (\pi_n Y)_\pi \rightarrow H_n Y \rightarrow H_n \pi \rightarrow 0.$$

Here $(\pi_n Y)_\pi$ makes sense because π acts on $\pi_n Y$, cf. Spanier [1966], ch. 7, sec. 3; this action corresponds to the obvious action of π on $H_n X$ (X = universal cover of Y) under the isomorphism $\pi_n Y \cong \pi_n X \cong H_n X$. [Remark: Without any dimension restrictions on Y one can prove that there is an exact sequence

$$H_{n+1} Y \rightarrow H_{n+1} \pi \rightarrow (\pi_n Y)_\pi \rightarrow H_n Y \rightarrow H_n \pi \rightarrow 0.$$

This will be proved in exercise 6 of §VII.7 by a spectral sequence argument; the reader may want to try to give a direct proof now.]

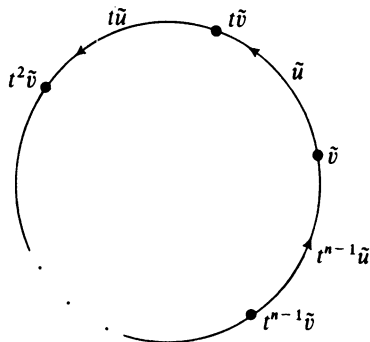
2. Let $G = \langle S; r_1, r_2, \dots \rangle$, i.e., $G = F(S)/R$ where R is the normal closure in $F(S)$ of $r_1, r_2, \dots$.

(a) Show that the relation module R_{ab} is generated (as G -module) by the images of $r_1, r_2, \dots$.

(b) Show that these elements *freely* generate R_{ab} as $\mathbb{Z}G$ -module if and only if the 2-complex associated to the given presentation of G is a $K(G, 1)$. (By the 2-complex associated to the presentation we mean the complex $(\bigvee_{s \in S} S_s^1) \bigcup_{r_1} e_1^2 \bigcup_{r_2} e_2^2 \cup \dots$, where the 2-cell e_i^2 is attached to $\bigvee S_s^1$ by the map $S^1 \rightarrow \bigvee S_s^1$ corresponding to $r_i \in \pi_1(\bigvee S_s^1) = F(S)$.) Thus, for example, Lyndon's theorem about one-relator groups $G = \langle S; r \rangle$ which we stated in example 3 of §4 can be interpreted as saying that R_{ab} is freely generated by the image of r , provided r is not a power. (It is in this form, in fact, that Lyndon stated and proved his theorem.)

*(c) Let $G = \langle S; r \rangle$ be an *arbitrary* one-relator group and write $r = u^n$ in $F(S)$, where $n \geq 1$ is maximal. One can show (cf. Lyndon-Schupp [1977], IV.5.2) that the image t of u in G has order exactly n , and we denote by C the cyclic group of order n generated by t . If $n > 1$ then R_{ab} is not freely generated by $r \bmod [R, R]$, since this generator is clearly fixed by C . But Lyndon [1950] proved that no other relations hold, i.e., that the obvious surjection $\mathbb{Z}[G/C] \rightarrow R_{ab}$ is an isomorphism. Show that this result can be interpreted topologically, as follows. Let Y , $\tilde{Y}$, and $\tilde{v}$ be as in the

proof of 5.3 and consider the lifting $\tilde{r}$ of the loop r in Y . Since $r = u^n$ and $\tilde{u}$ ends at $\tilde{v}$, $\tilde{r}$ is the composite path pictured schematically as follows:



Thus the map $S^1 \rightarrow \tilde{Y}$ corresponding to $\tilde{r}$ is compatible with the action of the cyclic group C , where C acts on S^1 as a group of rotations as in §I.6. Deduce that we can form a 2-dimensional G -complex X by attaching 2-cells to $\tilde{Y}$ along the loops $g\tilde{r}$, where g ranges over a set of representatives for the cosets G/C ; if σ is the 2-cell attached along $\tilde{r}$, then the isotropy group G_σ is equal to C , with C acting on σ as a group of rotations. Show that Lyndon's theorem about R_{ab} stated above is equivalent to the statement that X is contractible. [Note: In the terminology of Lyndon-Schupp [1977], ch. III, X is the *Cayley complex* associated to the presentation $\langle S; r \rangle$.]

3. In this exercise you will construct the Fox "free derivatives" by using the ideas introduced in the proof of 5.3. See the exercises in §IV.2 for a purely algebraic treatment of the same results. If G is a group and M a G -module, then a *derivation* (or *crossed homomorphism*) from G to M is a function $d: G \rightarrow M$ such that $d(gh) = dg + gdh$ for all $g, h \in G$.

(a) Let the notation be as in 5.3 and its proof. Show that the definition of dr given in that proof can be used (almost verbatim) to define a function $d: F \rightarrow C_1 \tilde{Y}$ which satisfies $d(f_1 f_2) = df_1 + f_1 df_2$ for all $f_1, f_2 \in F$. Thus if we regard the G -module $C_1 \tilde{Y}$ as an F -module via the canonical homomorphism $F \rightarrow G$, then $d: F \rightarrow C_1 \tilde{Y}$ is a derivation.

(b) For any free group $F = F(S)$, show that there is an F -module Ω which admits a derivation $d: F \rightarrow \Omega$ such that Ω is a free $\mathbb{Z}F$ -module with basis $(ds)_{s \in S}$. [Hint: Apply (a) with $R = \{1\}$.] We call df for $f \in F$ the *total (free) derivative* of f . The coefficient of ds when df is expressed in terms of the basis (ds) is called the *partial derivative* of f with respect to s and is denoted $\partial f / \partial s$; thus $df = \sum_{s \in S} (\partial f / \partial s) ds$, where $\partial f / \partial s \in \mathbb{Z}F$. Show that $\partial / \partial s: F \rightarrow \mathbb{Z}F$ is a derivation and satisfies $\partial t / \partial s = \delta_{s,t}$ ($t \in S$). These properties completely characterize $\partial / \partial s$. (For example, if $S = \{s, t\}$, you should be able to use these properties to compute $\partial(ts^{-1}ts^2) / \partial s$.)

(c) With F as in (b), show that any derivation $d: F \rightarrow M$, where M is an F -module, satisfies $df = \sum_{s \in S} (\partial f / \partial s) ds$.

(d) Show that the map $\theta: R_{ab} \rightarrow \mathbb{Z}G^{(S)}$ in 5.4 is induced by the map $R \rightarrow \mathbb{Z}G^{(S)}$ given by $r \mapsto \sum_{s \in S} (\partial r / \partial s) e_s$ where $\partial r / \partial s$ is the image of $\partial r / \partial s$ under the canonical map

$\mathbb{Z}F \rightarrow \mathbb{Z}G$. If R is the normal closure of a subset $T \subseteq F$, deduce that there is a partial free resolution

$$\mathbb{Z}G^{(T)} \xrightarrow{\partial_2} \mathbb{Z}G^{(S)} \xrightarrow{\partial_1} \mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

such that the matrix of ∂_2 is the "Jacobian matrix" $(\partial t / \partial s)_{t \in T, s \in S}$.

- *4. Let $G = F/R$ as in 5.3. The purpose of this exercise is to establish explicit formulas for the isomorphism $\varphi: H_2 G \xrightarrow{\sim} R \cap [F, F]/[F, R]$ and its inverse, where we view $H_2 G$ as the second homology group of the standard chain complex $C_*(G)$.

(a) Exhibit a specific chain map in dimensions ≤ 2 from the bar resolution to the partial resolution

$$(*) \quad \mathbb{Z}G^{(R)} \rightarrow C_1 \tilde{Y} \rightarrow C_0 \tilde{Y} \rightarrow \mathbb{Z} \rightarrow 0,$$

where $\tilde{Y}$ is as in the proof of 5.3. (Here $\mathbb{Z}G^{(R)}$ is a free module with basis $(e_r)_{r \in R}$; it maps onto $R_{ab} = H_1 \tilde{Y} \subset C_1 \tilde{Y}$ in the obvious way.) Deduce that φ can be computed as follows: Choose for each $g \in G$ an element $\bar{f}(g) \in F$ such that $\overline{\bar{f}(g)} = g$. Given $g, h \in G$, write $\bar{f}(g)\bar{f}(h) = \bar{f}(gh)r(g, h)$ where $r(g, h) \in R$. Then there is an abelian group homomorphism $C_2(G) \rightarrow R_{ab}$ given by $[g|h] \mapsto r(g, h) \bmod [R, R]$, and this induces the isomorphism $\varphi: H_2 G \rightarrow R \cap [F, F]/[F, R]$ by passage to subquotients.

(b) Exhibit a chain map from $(*)$ to the bar resolution, and deduce that φ^{-1} can be computed as follows: Let $D: F \rightarrow C_2(G)$ be the unique derivation such that $Ds = [1|\bar{s}]$ for each free generator $s \in S$, where the group $C_2(G)$ is regarded as an F -module by $f \cdot [g|h] = [\bar{f}g|h]$. (Explicitly, $Df = \sum_{s \in S} [\partial f / \partial s | \bar{s}]$, where the symbol $[\cdot | \cdot]$ is understood to be $\mathbb{Z}$ -bilinear.) Then $(D|R): R \rightarrow C_2(G)$ is a homomorphism which induces φ^{-1} by passage to subquotients.

(c) Let $a_1, \dots, a_g, b_1, \dots, b_g$ be elements of F such that the element $r = \prod_{i=1}^g [a_i, b_i]$ is in R . Then $\varphi^{-1}(r \bmod [F, R])$ is represented by the cycle

$$\sum_{i=1}^g \{ [I_{i-1} | \bar{a}_i] + [I_{i-1} \bar{a}_i | \bar{b}_i] - [I_{i-1} \bar{a}_i \bar{b}_i \bar{a}_i^{-1} | \bar{a}_i] - [I_i | \bar{b}_i] \},$$

where $I_i = [\bar{a}_1, \bar{b}_1] \cdots [\bar{a}_i, \bar{b}_i]$. [Hint: It suffices to prove this in the universal example where F is the free group on $a_1, \dots, a_g, b_1, \dots, b_g$ and R is the normal closure of r . In this case, $\varphi^{-1}(r \bmod [F, R])$ is easily computed by the formula of (b).]

5. (a) Let G be a group which admits a presentation with n generators and m relations. Let $r = \text{rk}_{\mathbb{Z}}(G_{ab}) = \dim_{\mathbb{Q}}(\mathbb{Q} \otimes G_{ab})$. Prove that the abelian group $H_2 G$ can be generated by $m - n + r$ elements. [Hint: Let Y be the 2-complex associated to the presentation. Computing the Euler characteristic $\chi(Y)$ in two different ways, one finds $1 - n + m = 1 - \text{rk}(H_1 Y) + \text{rk}(H_2 Y) = 1 - r + \text{rk}(H_2 Y)$, whence $\text{rk}(H_2 Y) = m - n + r$. Now $H_2 Y$ is a free abelian group, being a subgroup of the group of cellular 2-chains of Y , and we have a surjection $H_2 Y \twoheadrightarrow H_2 G$.]

Parts (b) and (c) below illustrate typical applications of (a).

(b) Let G be a perfect group (i.e., a group such that $G_{ab} = 0$) which admits a finite presentation with the same number of generators as relations. Prove that $H_2 G = 0$.

(c) Let G be a perfect group such that $H_2 G = \mathbb{Z}_2 \oplus \mathbb{Z}_2$. Show that any n -generator presentation of G must involve at least $n + 2$ relations.

6. (a) By a *group extension* we mean a short exact sequence $1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1$ of groups. Deduce from Hopf's formula that such an extension gives rise to a 5-term exact sequence

$$H_2 G \xrightarrow{\alpha} H_2 Q \xrightarrow{\beta} (H_1 N)_Q \xrightarrow{\gamma} H_1 G \xrightarrow{\delta} H_1 Q \rightarrow 0,$$

where the Q -action on $H_1 N = N_{ab}$ is induced by the conjugation action of G on N , and γ and δ are induced by the maps $N \hookrightarrow G \rightarrow Q$. [Hint: Write $G = F/R$ and $Q = F/S$, where $R \subseteq S \subseteq F$. The desired sequence is then

$$\frac{R \cap [F, F]}{[F, R]} \rightarrow \frac{S \cap [F, F]}{[F, S]} \rightarrow \frac{N}{[G, N]} \rightarrow \frac{G}{[G, G]} \rightarrow \frac{G}{N \cdot [G, G]} \rightarrow 0,$$

which is easily proved to be exact.] Remark: We will give another derivation of this 5-term exact sequence, and a generalization of it, in §VII.6. It has interesting applications to the study of the lower central series, due independently to Stallings [1965a] and Stambach [1966]. See also Stambach [1973], Chapter IV, for further developments along these lines.

(b) Conversely, show that Hopf's formula can be deduced from the 5-term exact sequence. [Apply (a) to $1 \rightarrow R \rightarrow F \rightarrow G \rightarrow 1$ and recall that $H_2 F = 0$.]

- *7. (a) Recall that the 3-sphere S^3 has a group structure. [It is the multiplicative group of quaternions of norm 1.] If G is a finite subgroup of S^3 , deduce from the results of this section that $H_2 G = 0$. [Hint: S^3/G is a closed, orientable 3-manifold with finite fundamental group, hence $H_2(S^3/G) = 0$.] Remark: A list of the groups G to which this applies is given in Wolf [1974] and recalled briefly in example 2 of VI.9.2 below. We will also be able to give there a more elementary solution of the present exercise, cf. exercise 3 of §VI.9.

(b) One of the most interesting groups to which (a) applies is the "binary icosahedral group." This is a group G of order 120 which maps onto A_5 , the alternating group on 5 letters, with central kernel of order 2. Moreover, G is perfect. Using these facts, the result of (a), and the 5-term exact sequence, deduce that $H_2(A_5) = \mathbb{Z}_2$.

(c) (R. Strebél) It is known that A_5 admits the presentation $\langle x, y, z; x^2 = y^3 = z^5 = xyz = 1 \rangle$. Consider now the abstract group $G = \langle x, y, z; x^2 = y^3 = z^5 = xyz \rangle$, and let C be the cyclic subgroup of G generated by the central element $x^2 = y^3 = z^5 = xyz$ of G . Thus $G/C = A_5$. Determine the order of C and hence that of G . [Hint: From the given presentation of G you can show that G is perfect, hence $H_2 G = 0$ by exercise 5b above. The 5-term exact sequence now yields $H_2(A_5) \xrightarrow{\cong} H_1(C) = C$, so C has order 2 and hence G has order 120. In fact, with a little more work you can show that G is the binary icosahedral group.] Similar methods can be used to analyze other abstract groups defined by presentations closely related to presentations of known groups. You might look, for instance, at some of the examples in Coxeter-Moser [1980] in connection with Miller's generalization of the polyhedral groups; the treatment there can be simplified by the use of the method of the present exercise.

Remark. It is clear from this exercise that H_2 is closely related to the theory of central extensions. It would be possible to develop this connection systematically on the basis of Hopf's formula and the 5-term exact sequence, but we will instead deduce it from the general theory of group extensions, to be discussed in Chapter IV. See exercise 7 in §IV.3.

6 Functoriality

The standard chain complex $C_*(G)$ is clearly functorial in G , hence H_*G is a (covariant) functor of G . This functoriality can also be described in terms of arbitrary resolutions, as follows: Given a homomorphism $\alpha: G \rightarrow G'$ and projective resolutions F and F' of $\mathbb{Z}$ over $\mathbb{Z}G$ and $\mathbb{Z}G'$, respectively, we can regard F' as a complex of G -modules via α . Then F' is acyclic (although not projective, in general, over $\mathbb{Z}G$), so the fundamental lemma (I.7.4) gives us an augmentation-preserving G -chain map $\tau: F \rightarrow F'$, well-defined up to homotopy. The condition that τ be a G -map is expressed by the formula

$$(6.1) \quad \tau(gx) = \alpha(g)\tau(x)$$

for $g \in G, x \in F$. Clearly τ induces a map $F_G \rightarrow F'_{G'}$, well-defined up to homotopy, hence we obtain a well-defined map $\alpha_*: H_*G \rightarrow H_*G'$.

(6.2) Proposition. Fix $g_0 \in G$ and let $\alpha: G \rightarrow G$ be given by $\alpha(g) = g_0 g g_0^{-1}$. Then $\alpha_*: H_*G \rightarrow H_*G$ is the identity.

PROOF. Let F be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ and define $\tau: F \rightarrow F$ by $\tau(x) = g_0 x$. Then τ commutes with the boundary operator and satisfies 6.1, so τ can be used to compute α_* . But clearly τ induces the identity map on F_G , whence the proposition. $\square$

(6.3) Corollary. If G is a group and N is a normal subgroup, then the conjugation action of G on N induces an action of G/N on H_*N .

EXERCISES

- Given $N \triangleleft G$ as in 6.3, let F be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ and consider the complex F_N . Then F_N is a complex of G/N -modules (§2, exercise 3a), hence $H_*(F_N)$ inherits an action of G/N . Show that $H_*(F_N) = H_*N$ and that the resulting G/N -action on H_*N agrees with that defined in 6.3. [Hint: Given $g \in G$, the action of g on H_*N can be computed via the map $\tau: F \rightarrow F$ given by $\tau(x) = gx$.]
- For any finite set A let $\Sigma(A)$ be the group of permutations of A . If $|A| \leq |B|$ (where $|\cdot|$ denotes cardinality), choose an injection $i: A \hookrightarrow B$ and consider the injection $\Sigma(A) \hookrightarrow \Sigma(B)$ obtained by extending a permutation of A to be the identity on $B - iA$. Show that the induced map $H_*\Sigma(A) \rightarrow H_*\Sigma(B)$ is independent of the choice of i . In particular, if $|A| = |B|$, then $H_*\Sigma(A)$ is canonically isomorphic to $H_*\Sigma(B)$.
- (a) Under the isomorphism $H_1(\) \approx (\)_{ab}$ of §3, show that $H_1(\alpha): H_1(G) \rightarrow H_1(G')$ corresponds to the map $G_{ab} \rightarrow G'_{ab}$ obtained from α by passage to the quotient; in other words, the isomorphism $H_1(\) \approx (\)_{ab}$ is natural. In particular, the action of G/N on $H_1(N)$ in 6.3 above agrees with that defined in exercise 6 of §5. [Hint: Use the bar resolution to compute $H_1(\alpha)$.]

(b) Prove the following naturality property of Hopf's isomorphism 5.3: Suppose $G = F/R$ and $G' = F'/R'$ with F and F' free, and suppose $\alpha: G \rightarrow G'$ lifts to $\tilde{\alpha}: F \rightarrow F'$. Then the diagram

$$\begin{array}{ccc} H_2(G) \approx R \cap [F, F]/[F, R] & & \\ \downarrow H_2(\alpha) & & \downarrow \\ H_2(G') \approx R' \cap [F', F']/[F', R'] & & \end{array}$$

commutes, where the right-hand vertical arrow is induced by $\tilde{\alpha}$. [Hint: Let Y and $\tilde{Y}$ be associated to the presentation $G = F/R$ as in the proof of 5.3, and similarly let Y' and $\tilde{Y}'$ correspond to $G' = F'/R'$. Then $\tilde{\alpha}$ yields a map $Y \rightarrow Y'$, which yields a map $C_*(\tilde{Y}) \rightarrow C_*(\tilde{Y}')$, which can be extended to a map τ of resolutions, which can be used to compute $H_2(\alpha)$.]

Remark. This exercise allows one to interpret in terms of the functoriality of H_1 and H_2 three of the four maps which occur in the 5-term exact sequence of exercise 6 of §5.

7 The Homology of Amalgamated Free Products

As an illustration of the topological interpretation of group homology, we will derive in this section a Mayer–Vietoris sequence for computing the homology of an amalgamated free product. We begin by reviewing the necessary group theory.

Suppose we are given groups G_1 , G_2 , and A and homomorphisms $\alpha_1: A \rightarrow G_1$ and $\alpha_2: A \rightarrow G_2$. Eventually we will assume further that α_1 and α_2 are injective, so that A can be viewed as a common subgroup of G_1 and G_2 , but for the moment we do not make this assumption. By the *amalgamated free product* (or *amalgamated sum*, or *amalgam*) of G_1 and G_2 along A we mean a group G which fits into a commutative square

$$(7.1) \quad \begin{array}{ccc} A & \xrightarrow{\alpha_2} & G_2 \\ \alpha_1 \downarrow & & \downarrow \beta_2 \\ G_1 & \xrightarrow{\beta_1} & G \end{array}$$

with the following *universal mapping property*: Given a group H and homomorphisms $\gamma_i: G_i \rightarrow H$ ($i = 1, 2$) with $\gamma_1\alpha_1 = \gamma_2\alpha_2$, there is a unique map $\varphi: G \rightarrow H$ such that $\varphi\beta_i = \gamma_i$. We write $G = G_1 *_A G_2$, and we say that the square 7.1 is an *amalgamation diagram*.

The universal property above shows that amalgamation is the group-theoretic analogue of pasting two topological spaces together along a common subspace. The Seifert–van Kampen theorem, which the reader has probably seen in some form, makes the analogy precise via the π_1 -functor. We will need the following simple version of that theorem:

(7.2) Theorem. *Let X be a CW-complex which is the union of two connected subcomplexes X_1 and X_2 whose intersection Y is connected and non-empty. Then the square*

$$\begin{array}{ccc} \pi_1 Y & \longrightarrow & \pi_1 X_2 \\ \downarrow & & \downarrow \\ \pi_1 X_1 & \longrightarrow & \pi_1 X \end{array}$$

is an amalgamation diagram, where all fundamental groups are computed at a fixed vertex $y \in Y$ and all maps are induced by inclusions. Thus

$$\pi_1 X = \pi_1 X_1 *_{\pi_1 Y} \pi_1 X_2.$$

This is an easy consequence of the usual combinatorial description of the fundamental group of a CW-complex. Details can be found in Schubert [1968], III.5.8, or Cohen [1978], II.2.3. See also exercise 2 below for a proof based on covering space theory.

We can express this theorem more concisely by saying that the functor

$$\pi_1: (\text{connected, pointed complexes}) \rightarrow (\text{groups})$$

preserves amalgamations. In order to study the homology of amalgamations of groups, we would like to have a result going in the other direction, saying that the “functor” $K(-, 1): (\text{groups}) \rightarrow (\text{complexes})$ preserves amalgamations. This turns out to be true as long as the maps α_1 and α_2 of 7.1 are injective:

(7.3) Theorem (Whitehead). *Any amalgamation diagram 7.1 with α_1 and α_2 injective can be realized by a diagram*

$$\begin{array}{ccc} Y & \hookrightarrow & X_2 \\ \downarrow & & \downarrow \\ X_1 & \hookrightarrow & X \end{array}$$

of $K(\pi, 1)$ -complexes such that $X = X_1 \cup X_2$ and $Y = X_1 \cap X_2$.

The proof will require three elementary lemmas:

(7.4) Lemma. *If α_1 and α_2 are injective then so are β_1 and β_2 . Thus G_1 , G_2 , and A can be regarded as subgroups of G .*

This is part of the “normal form theorem” for amalgamations. See, for instance, Serre [1977a], Lyndon–Schupp [1977], or Cohen [1978].

(7.5) Lemma. *Let $X' \hookrightarrow X$ be an inclusion of connected CW-complexes such that the induced map $\pi' \rightarrow \pi$ of fundamental groups is injective. Let $p: \tilde{X} \rightarrow X$ be the universal cover of X . Then each connected component of $p^{-1}X'$ is simply*

connected (hence it is a copy of the universal cover of X'). Moreover, these components are permuted transitively by the action of π on $\tilde{X}$, and π' is the isotropy group of one of them; in other words, $\pi_0(p^{-1}X') \approx \pi/\pi'$.

PROOF. For any basepoint in $p^{-1}X'$ we have a diagram

$$\begin{array}{ccc} \pi_1(p^{-1}X') & \longrightarrow & \pi_1\tilde{X} \\ \downarrow & & \downarrow \\ \pi' & \hookrightarrow & \pi, \end{array}$$

where the vertical maps are induced by p and the horizontal maps by inclusions. Since $\pi_1\tilde{X} = \{1\}$, the first assertion follows at once. The second assertion, which we will not make serious use of, is left as an exercise for the interested reader. $\square$

(7.6) Lemma. Any diagram $G_1 \leftarrow A \rightarrow G_2$ of groups can be realized by a diagram $X_1 \leftarrow Y \hookrightarrow X_2$ of $K(\pi, 1)$ -complexes.

PROOF. According to exercise 4 of §I.4 or exercise 3 of §I.5, $K(\pi, 1)$ -complexes can be constructed functorially. We can therefore realize the group homomorphisms by cellular maps $X_1 \leftarrow Y \rightarrow X_2$ of $K(\pi, 1)$'s. Taking mapping cylinders if necessary (cf. Spanier [1966], 1.4), we can make these maps inclusions. $\square$

PROOF OF 7.3. Start with $X_1 \leftarrow Y \hookrightarrow X_2$ as in 7.6 and form the adjunction space $X = X_1 \cup_Y X_2$, i.e., X is obtained from the disjoint union $X_1 \amalg X_2$ by identifying the two copies of Y . Then $\pi_1 X = G_1 *_A G_2 = G$ by 7.2, so we need only show that the universal cover $\tilde{X}$ satisfies $H_i \tilde{X} = 0$ for $i > 1$. Let $\tilde{X}_1, \tilde{X}_2$, and $\tilde{Y}$ be the inverse images of X_1, X_2 , and Y in $\tilde{X}$. Since X_1, X_2 , and Y have acyclic universal covers, it follows from 7.4 and 7.5 that $\tilde{X}_1, \tilde{X}_2$, and $\tilde{Y}$ have trivial homology in positive dimensions. The Mayer-Vietoris sequence associated to the square

$$\begin{array}{ccc} \tilde{Y} & \hookrightarrow & \tilde{X}_2 \\ \downarrow & & \downarrow \\ \tilde{X}_1 & \hookrightarrow & \tilde{X} \end{array}$$

therefore shows that $H_i \tilde{X} = 0$ for $i > 1$. $\square$

(7.7) Corollary. Given $G = G_1 *_A G_2$ where $A \hookrightarrow G_1$ and $A \hookrightarrow G_2$, there is a "Mayer-Vietoris" sequence

$$\cdots \rightarrow H_n A \rightarrow H_n G_1 \oplus H_n G_2 \rightarrow H_n G \rightarrow H_{n-1} A \rightarrow \cdots.$$

This is immediate from 7.3.

Remark. As a bi-product of the proof of 7.3 we obtain an exact sequence of permutation modules

$$(7.8) \quad 0 \rightarrow \mathbb{Z}[G/A] \rightarrow \mathbb{Z}[G/G_1] \oplus \mathbb{Z}[G/G_2] \rightarrow \mathbb{Z} \rightarrow 0.$$

Indeed, this is just the low-dimensional part of the Mayer–Vietoris sequence which we used in the proof of 7.3. (The second part of 7.5 allows one to identify $H_0(\tilde{Y})$, $H_0(\tilde{X}_1)$, and $H_0(\tilde{X}_2)$ with permutation modules.) We will give another derivation of 7.8, again based on topological ideas, in an appendix to this chapter. It is also possible to prove 7.8 algebraically, cf. Bieri [1976], Prop. 2.8, or Swan [1969], Lemma 2.1. Moreover, it is possible to give a purely algebraic proof of 7.7, using 7.8 as the starting point. We will explain this in the exercise of §III.6 below.

EXERCISES

1. Let $G = G_1 *_A G_2$ with $A \rightarrow G_1$ and $A \rightarrow G_2$ not necessarily injective. Let $\bar{G}_1, \bar{G}_2$ and $\bar{A}$ be the images of G_1, G_2 , and A in G . Show that $G = \bar{G}_1 *_A \bar{G}_2$. Thus any amalgam as defined at the beginning of this section is isomorphic to one in which the maps $A \rightarrow G_1$ and $A \rightarrow G_2$ are injective.
- *2. Give a proof of 7.2 based on the classification of regular covering spaces, as stated in the appendix to Chapter I, A2. [Hint: For any group H , a pointed regular H -covering $\tilde{X}$ of X is specified by giving pointed regular H -coverings $\tilde{X}_1$ of X_1 and $\tilde{X}_2$ of X_2 such that the induced coverings of Y are isomorphic (as pointed regular H -coverings).]
3. It is a classical fact that $SL_2(\mathbb{Z}) \approx \mathbb{Z}_4 *_2 \mathbb{Z}_6$. (See Serre [1977a] for an indication of an easy proof of this; see also example 3 of §VIII.9 below.) Use the Mayer–Vietoris sequence to calculate $H_*(SL_2(\mathbb{Z}))$. [Suggestion: You can save a lot of work by considering separately the 2-torsion and the 3-torsion in the Mayer–Vietoris sequence. As far as 2-torsion is concerned, $SL_2(\mathbb{Z})$ behaves like $\mathbb{Z}_4 *_2 \mathbb{Z}_2 = \mathbb{Z}_4$, whereas it behaves like $\{1\} *_1 \mathbb{Z}_3 = \mathbb{Z}_3$ with respect to 3-torsion.]

Appendix. Trees and Amalgamations

The results of §7 were based on a topological interpretation of amalgamated free products in terms of “amalgamations” of topological spaces. The purpose of this appendix is to describe a different topological interpretation of amalgamated free products, due to Serre [1977a]. In particular, we will obtain another proof of 7.8.

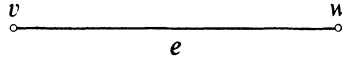
Recall that a *graph* is a 1-dimensional CW-complex and that a *path* in a graph is a sequence $e_1, \dots, e_n$ of oriented edges such that the final vertex of e_i equals the initial vertex of e_{i+1} for $1 \leq i < n$. The path is a *loop* if the final vertex of e_n equals the initial vertex of e_1 . We allow the case $n = 0$, in which

case the path is called *trivial*. Finally, the path is called *reduced* if $e_{i+1} \neq \bar{e}_i$ for $0 \leq i < n$, where $\bar{e}_i$ is the same geometric edge as e_i but with the opposite orientation.

The graph X is called a *tree* if it satisfies the following conditions, which are easily seen to be equivalent:

- (i) X is contractible.
- (ii) X is simply connected.
- (iii) X is acyclic.
- (iv) X is connected and contains no non-trivial reduced loops.

Suppose a group G acts as a group of automorphisms of a tree X , and let e be an edge of X with vertices v and w . We will say that e is a *fundamental domain* for the G -action if every edge of X is equivalent to $e \bmod G$ and every vertex of X is equivalent to either v or w but not both. In other words, we require that the subgraph



map isomorphically onto the orbit graph X/G . Note, in this case, that the isotropy groups G_e , G_v , and G_w satisfy

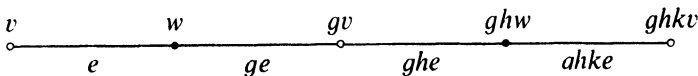
$$G_e = G_v \cap G_w.$$

Indeed, we have $G_e \supseteq G_v \cap G_w$ since e is the only edge with vertices v and w [X is a tree]; and the opposite inclusion holds since no element of G can interchange v and w [they are inequivalent mod G].

The theorem of Serre that we wish to state says that actions of G of this type (i.e., with an edge as fundamental domain) are essentially the same as decompositions of G as an amalgamated free product:

(A1) Theorem. *Let a group G act on a tree X . Suppose e is an edge with vertices v and w such that e is a fundamental domain for the action. Then $G = G_v *__{G_e} G_w$. Conversely, given an amalgamation $G = G_1 *_A G_2$ (where $A \hookrightarrow G_1$ and $A \hookrightarrow G_2$), there exists a tree on which G acts as above, with G_1 , G_2 , and A as the isotropy groups G_v , G_w , and G_e .*

SKETCH OF PROOF. Note that two edges $g_1 e$ and $g_2 e$ of X ($g_i \in G$) have a vertex in common if and only if $g_1^{-1} g_2 \in G_v$ or G_w . This allows one to relate reduced paths in X to reduced words ("normal forms") in $G_v *__{G_e} G_w$. Consider, for example, a reduced path of length 4 starting with e , oriented so that v is its first vertex. Such a path must have the form



where $g, k \in G_w$ and $h \in G_v$. Moreover, $g, h, k \notin G_e$. Thus the path gives rise to the reduced word ghk in $G_v *__{G_e} G_w$. Since X has no loops, we must have

$ghk \neq 1$ in G . Generalizing this argument, one sees that the canonical map $\varphi: G_v *_G G_w \rightarrow G$ is injective. Similarly, it follows from the connectivity of X that φ is surjective. [Given $g \in G$, consider a path ending at gv , with e as its first edge.] This proves the first part of the theorem.

Conversely, given $G = G_1 *_A G_2$, we have no choice as to how to construct X . Namely, we must take $G/G_1 \amalg G/G_2$ as the set of vertices of X and G/A as the set of edges. There are canonical maps $\alpha: G/A \rightarrow G/G_1$ and $\beta: G/A \rightarrow G/G_2$, by which we attach the edges to the vertices, i.e., an edge $e \in G/A$ joins the vertices $\alpha(e)$ and $\beta(e)$. In this way we obtain a graph X on which G acts with an edge as fundamental domain and with G_1 , G_2 , and A as the isotropy groups. Since $G = G_1 *_A G_2$, the first part of the proof shows that X is connected and has no non-trivial reduced loops, so that it is a tree. $\square$

It is apparent from this sketch of the proof that the theorem is not particularly deep; indeed, the existence of the tree X associated to $G_1 *_A G_2$ is little more than a reformulation of the normal form theorem for amalgamated free products. Nevertheless, the tree is a very convenient tool for keeping track of the combinatorics of normal forms. It is often considerably easier to prove things about G by using X than it is to work directly with normal forms.

For example, we obtain the promised proof of 7.8 by noting that the sequence in 7.8 is simply the augmented chain complex of X . Its exactness therefore follows from the acyclicity of X .

Here are two other applications of A1:

(A2) Corollary. *Let $F \subseteq G_1 *_A G_2$ be a subgroup which intersects every conjugate of G_1 and G_2 trivially. Then F is a free group.*

PROOF. The hypothesis implies that F acts freely on X . The latter being simply connected, it follows that X is the universal cover of the orbit graph X/F and $F \approx \pi_1(X/F)$. But it is well-known that the fundamental group of a graph is free. $\square$

(A3) Corollary. *Let $H \subseteq G_1 *_A G_2$ be a finite subgroup. Then H is conjugate to a subgroup of G_1 or G_2 .*

PROOF. We must show that H fixes some vertex of X . But this follows from the elementary fact that every finite group of automorphisms of a tree has a fixed point. See Serre [1977a], I.4.3, for more details. $\square$

CHAPTER III

Homology and Cohomology with Coefficients

0 Preliminaries on $\otimes_G$ and Hom_G

Recall that the tensor product $M \otimes_R N$ is defined whenever M is a right R -module and N is a left R -module. It is the quotient of $M \otimes_{\mathbb{Z}} N$ (which we denote $M \otimes N$) obtained by introducing the relations $mr \otimes n = m \otimes rn$ ($m \in M, r \in R, n \in N$).

In case R is a group ring $\mathbb{Z}G$, we can avoid having to consider both left and right modules by using the anti-automorphism $g \mapsto g^{-1}$ of G . Thus we can regard any left G -module M as a right G -module by setting $mg = g^{-1}m$ ($m \in M, g \in G$), and in this way we can make sense out of the tensor product $M \otimes_{\mathbb{Z}G} N$ (also denoted $M \otimes_G N$) of two left G -modules.

Note that $M \otimes_G N$ is obtained from $M \otimes N$ by introducing the relations $g^{-1}m \otimes n = m \otimes gn$. If we replace m by gm , these relations take the form $m \otimes n = gm \otimes gn$, and we see that

$$(0.1) \quad M \otimes_G N = (M \otimes N)_G,$$

where G acts “diagonally” on $M \otimes N$: $g \cdot (m \otimes n) = gm \otimes gn$. In particular, this shows that $-\otimes_G-$ is commutative:

$$M \otimes_G N \approx N \otimes_G M.$$

Warning. The passage between left and right modules as above is convenient, but it can sometimes be confusing, for instance if M naturally admits both a left and a right G -action (e.g., $M = \mathbb{Z}G'$, where $G \subseteq G'$). In such cases we will revert to the standard notation $M \otimes_{\mathbb{Z}G} N$ if we want to indicate that the tensor product is to be formed with respect to the given right action of G on M rather than the right action obtained from the left action.

The diagonal G -action used above is quite general and can be used whenever a functor of one or several abelian groups is applied to G -modules. Consider, for example, the functor $\text{Hom}(_, _) = \text{Hom}_{\mathbb{Z}}(_, _)$. If M and N are G -modules, then the action of G on M and N induces by functoriality a “diagonal” action of G on $\text{Hom}(M, N)$, given by

$$(gu)(m) = g \cdot u(g^{-1}m)$$

for $g \in G, u \in \text{Hom}(M, N), m \in M$. [The use of g^{-1} here is needed because of the contravariance of Hom in the first variable. In effect, we compensate for the contravariance by converting M to a right module.]

Note that $gu = u$ if and only if u commutes with the action of g . Thus

$$(0.2) \quad \text{Hom}_G(M, N) = \text{Hom}(M, N)^G.$$

This observation has already occurred implicitly in exercise 5 of §I.8, where we used an averaging procedure to convert an arbitrary element of $\text{Hom}(M, N)$ to an element of $\text{Hom}(M, N)^G = \text{Hom}_G(M, N)$.

EXERCISES

1. Let F be a flat $\mathbb{Z}G$ -module and M a G -module which is $\mathbb{Z}$ -torsion-free (i.e., $\mathbb{Z}$ -flat). Show that $F \otimes M$ (with diagonal G -action) is $\mathbb{Z}G$ -flat. [Hint: $(F \otimes M) \otimes_G - = (F \otimes M \otimes -)_G = F \otimes_G (M \otimes -)$, which is an exact functor.]
2. Let F be a projective $\mathbb{Z}G$ -module and M a $\mathbb{Z}$ -free G -module. Show that $F \otimes M$ (with diagonal G -action) is projective. [Hint: $\text{Hom}_G(F \otimes M, -) = \text{Hom}(F \otimes M, -)^G = \text{Hom}(F, \text{Hom}(M, -))^G = \text{Hom}_G(F, \text{Hom}(M, -))$, which is an exact functor. See exercise 3 of §5 below for an alternative proof.]

1 Definition of $H_*(G, M)$ and $H^*(G, M)$

Let F be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ and let M be a G -module. We define the *homology of G with coefficients in M* by

$$(1.1) \quad H_*(G, M) = H_*(F \otimes_G M).$$

Here $F \otimes_G M$ can be thought of as the complex obtained from F by applying the functor $- \otimes_G M$. Thus 1.1 is a natural generalization of the definition of $H_* G$ in Chapter II; indeed, we recover the latter by taking $M = \mathbb{Z}$:

$$(1.2) \quad H_*(G, \mathbb{Z}) = H_* G.$$

As in Chapter II, $H_*(G, M)$ is well-defined up to canonical isomorphism.

The complex $F \otimes_G M$ can also be thought of as the tensor product of chain complexes (cf. §I.0), where M is regarded as a chain complex concentrated in dimension 0. From this point of view there is a certain asymmetry

in 1.1 which did not appear in the context of Chapter II. A more symmetric looking definition is obtained by choosing projective resolutions of *both* $\mathbb{Z}$ and M , say $\varepsilon: F \rightarrow \mathbb{Z}$ and $\eta: P \rightarrow M$, and setting

$$(1.3) \quad H_*(G, M) = H_*(F \otimes_G P).$$

Fortunately, 1.3 is consistent with 1.1; for η induces a weak equivalence $F \otimes \eta: F \otimes_G P \rightarrow F \otimes_G M$ by I.8.6.

Note that I.8.6 also gives us a weak equivalence $\varepsilon \otimes P: F \otimes_G P \rightarrow \mathbb{Z} \otimes_G P$. Thus

$$(1.4) \quad H_*(G, M) = H_*(P_G).$$

Clearly, then, we have considerable flexibility in the choice of a chain complex from which to compute $H_*(G, M)$. For the moment we content ourselves with a trivial example of such a computation:

$$(1.5) \quad H_0(G, M) \approx M_G.$$

This follows from the right exactness of the tensor product. [Apply $-\otimes_G M$ to $F_1 \rightarrow F_0 \rightarrow \mathbb{Z} \rightarrow 0$ and use 1.1, or apply $(\)_G$ to $P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$ and use 1.4.]

We turn now to *cohomology with coefficients*, which is defined via $\mathcal{H}om$ rather than $\otimes$. Choose a projective resolution $F \rightarrow \mathbb{Z}$ as above and consider the complex $\mathcal{H}om_G(F, M)$, where M is again regarded as a chain complex concentrated in dimension 0. Checking the definition of $\mathcal{H}om$ in §I.0, we see that $\mathcal{H}om_G(F, M)_n = \text{Hom}_G(F_{-n}, M)$. It is therefore reasonable to regard $\mathcal{H}om_G(F, M)$ as a *cochain complex* by the usual indexing conventions, i.e., by setting

$$\mathcal{H}om_G(F, M)^n = \mathcal{H}om_G(F, M)_{-n} = \text{Hom}_G(F_n, M).$$

It is then a non-negative cochain complex with coboundary operator δ given by

$$(\delta u)(x) = (-1)^{n+1} u(\partial x)$$

for $u \in \mathcal{H}om_G(F, M)^n$, $x \in F_{n+1}$. This formula is most easily remembered in the form

$$(1.6) \quad \langle \delta u, x \rangle + (-1)^{\deg u} \langle u, \partial x \rangle = 0,$$

cf. exercise 3 of §I.0.

We now define

$$(1.7) \quad H^*(G, M) = H^*(\mathcal{H}om_G(F, M)).$$

Remark. Because of the sign in the definition of δ above, $\mathcal{H}om_G(F, M)$ is *not* the same as the complex $\text{Hom}_G(F, M)$ obtained from F by applying the contravariant functor $\text{Hom}_G(-, M)$ dimension-wise. This somewhat annoying fact of nature is not serious, for changing the sign of a coboundary

operator does not change cocycles, coboundaries, or cohomology. Our convention of using $\mathcal{H}om$ instead of Hom is dictated by our desire to be able to use the properties of $\mathcal{H}om$ developed systematically in the exercises of §I.0. The reader is warned, however, that $H^*(G, M)$ is often defined in the literature by means of Hom instead of $\mathcal{H}om$.

Note that the exact sequence $F_1 \rightarrow F_0 \rightarrow \mathbb{Z} \rightarrow 0$ yields an exact sequence $0 \rightarrow \text{Hom}_G(\mathbb{Z}, M) \rightarrow \text{Hom}_G(F_0, M) \rightarrow \text{Hom}_G(F_1, M)$. Since $\text{Hom}_G(\mathbb{Z}, M) = M^G$, this gives

$$(1.8) \quad H^0(G, M) = M^G.$$

There exist analogues for $H^*(G, _)$ of 1.3 and 1.4, but these involve the notion of *injective resolution*, which we have only mentioned briefly (cf. exercise 4 of §I.7). The interested reader can find these analogues in exercise 4b below, which will not be needed elsewhere in this book.

EXAMPLES

1. Suppose G is infinite cyclic with generator t . Then we have a resolution (I.4.5)

$$0 \rightarrow \mathbb{Z}G \xrightarrow{t-1} \mathbb{Z}G \rightarrow \mathbb{Z} \rightarrow 0,$$

hence $H_*(G, M)$ is the homology of

$$\cdots \rightarrow 0 \rightarrow M \xrightarrow{t-1} M$$

and $H^*(G, M)$ is the cohomology of

$$M \xrightarrow{t-1} M \rightarrow 0 \rightarrow \cdots.$$

Thus $H_0(G, M) = H^1(G, M) = M_G$, $H_1(G, M) = H^0(G, M) = M^G$, and $H_i(G, M) = H^i(G, M) = 0$ for $i > 1$.

2. Suppose G is cyclic of finite order n with generator t . Then we have a resolution (I.6.3)

$$\cdots \xrightarrow{N} \mathbb{Z}G \xrightarrow{t-1} \mathbb{Z}G \xrightarrow{N} \mathbb{Z}G \xrightarrow{t-1} \mathbb{Z}G \rightarrow \mathbb{Z} \rightarrow 0,$$

where $N = \sum_{i=0}^{n-1} t^i$. Hence $H_*(G, M)$ is the homology of

$$\cdots \xrightarrow{N} M \xrightarrow{t-1} M \xrightarrow{N} M \xrightarrow{t-1} M$$

and $H^*(G, M)$ is the cohomology of

$$M \xrightarrow{t-1} M \xrightarrow{N} M \xrightarrow{t-1} M \xrightarrow{N} \cdots.$$

Note that $N: M \rightarrow M$ satisfies $Ngm = Nm$ ($g \in G, m \in M$) and that $NM \subseteq M^G$; thus N induces a map $\bar{N}: M_G \rightarrow M^G$, called the *norm map*. [This is, in fact, true for any finite group G , where $N = \sum_{g \in G} g$.] We can now read off from the above that

$$H_i(G, M) = H^{i+1}(G, M) = M^G/NM = \text{coker } \bar{N}$$

for i odd ($i \geq 1$) and that

$$H_i(G, M) = H^{i-1}(G, M) = \ker \bar{N}$$

for i even ($i \geq 2$).

3. For any group G we can always take our resolution F to be the bar resolution. In this case we write $C_*(G, M)$ for $F \otimes_G M [= M \otimes_G F]$ and $C^*(G, M)$ for $\mathcal{H}om_G(F, M)$. Thus an element of $C_n(G, M)$ can be uniquely expressed as a finite sum of elements of the form $m \otimes [g_1 | \dots | g_n]$, i.e., as a formal linear combination with coefficients in M of the symbols $[g_1 | \dots | g_n]$. The boundary operator $\partial: C_n(G, M) \rightarrow C_{n-1}(G, M)$ is given by

$$\begin{aligned} \partial(m \otimes [g_1 | \dots | g_n]) &= mg_1 \otimes [g_2 | \dots | g_n] \\ &\quad - m \otimes [g_1 g_2 | \dots | g_n] + \dots + (-1)^n m \otimes [g_1 | \dots | g_{n-1}]. \end{aligned}$$

Similarly, an element of $C^n(G, M)$ can be regarded as a function $f: G^n \rightarrow M$, i.e., as a function of n variables from G to M . The coboundary operator $\delta: C^{n-1}(G, M) \rightarrow C^n(G, M)$ is given, up to sign, by

$$\begin{aligned} (\delta f)(g_1, \dots, g_n) &= g_1 f(g_2, \dots, g_n) \\ &\quad - f(g_1 g_2, \dots, g_n) + \dots + (-1)^n f(g_1, \dots, g_{n-1}). \end{aligned}$$

(Note: If $n = 0$ then G^n is, by convention, a one-element set, so that $C^0(G, M) \approx M$.) We could also use the normalized bar resolution here; the resulting normalized cochain complex $C_N^*(G, M) \subseteq C^*(G, M)$ is the subcomplex consisting of those cochains f such that $f(g_1, \dots, g_n) = 0$ whenever some $g_i = 1$.

Finally, we remark that $H_*(G, M)$ and $H^*(G, M)$ have a topological interpretation analogous to that of $H_* G$:

$$H_*(G, M) = H_*(K(G, 1); \mathcal{M})$$

$$H^*(G, M) = H^*(K(G, 1); \mathcal{M}),$$

where $\mathcal{M}$ is the local coefficient system on $K(G, 1)$ associated to the G -module M . (See Eilenberg [1947], Chapter V, for the relevant facts about local coefficient systems.) We will not make use of these isomorphisms, so we leave the details to the interested reader. [Hint: Let F be the chain complex of the universal cover of a $K(G, 1)$ Y . Then $C_*(Y; \mathcal{M}) = F \otimes_G M$ and $C^*(Y; \mathcal{M}) = \mathcal{H}om_G(F, M)$.]

EXERCISES

1. Let G be a finite group, M a G -module, and $\bar{N}: M_G \rightarrow M^G$ the norm map defined in Example 2 above.
 - (a) Show that $\ker \bar{N}$ and $\text{coker } \bar{N}$ are annihilated by $|G|$. [Hint: Consider the obvious map $M^G \rightarrow M_G$.]

(b) Suppose M is a module of the form $M = \mathbb{Z}G \otimes A$, where A is an abelian group and G acts by $g \cdot (r \otimes a) = gr \otimes a$. [Such a module M is said to be *induced*.] Show that $\bar{N}: M_G \rightarrow M^G$ is an isomorphism.

(c) Show that $\bar{N}$ is an isomorphism if M is a projective $\mathbb{Z}G$ -module.

2. Use the standard cochain complex $C^*(G, M)$ to show that $H^1(G, M)$ is isomorphic to the group of derivations from G to M (§II.5, exercise 3) modulo the subgroup of principal derivations. (A principal derivation is one of the form $dg = gm - m$, where m is a fixed element of M .) In particular, if G acts trivially on M then $H^1(G, M) \approx \text{Hom}(G, M) = \text{Hom}(G_{ab}, M) = \text{Hom}(H_1 G, M)$.
3. Let A be an abelian group with trivial G -action. Show that $F \otimes_G A \approx F_G \otimes A$ and that $\mathcal{H}om_G(F, A) \approx \mathcal{H}om(F_G, A)$. Deduce universal coefficient sequences

$$0 \rightarrow H_n(G) \otimes A \rightarrow H_n(G, A) \rightarrow \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(G), A) \rightarrow 0$$

and

$$0 \rightarrow \text{Ext}_{\mathbb{Z}}^1(H_{n-1}(G), A) \rightarrow H^n(G, A) \rightarrow \text{Hom}(H_n(G), A) \rightarrow 0.$$

- *4. (a) State and prove an analogue of I.8.5 for maps into a non-negative cochain complex of injectives.
- (b) Let $\varepsilon: F \rightarrow \mathbb{Z}$ be a projective resolution and let $\eta: M \rightarrow Q$ be an injective resolution. Prove that ε and η induce weak equivalences

$$\mathcal{H}om_G(F, M) \rightarrow \mathcal{H}om_G(F, Q) \leftarrow \mathcal{H}om_G(\mathbb{Z}, Q),$$

so any of these three complexes can be used to compute $H^*(G, M)$. In particular,

$$H^*(G, M) \approx H^*(Q^G).$$

2 Tor and Ext

There are obvious generalizations of $H_*(G, —)$ and $H^*(G, —)$, called $\text{Tor}_*^G(—, —)$ and $\text{Ext}_*^G(—, —)$, obtained by removing the restriction in 1.1 and 1.7 that F be a resolution of the particular module $\mathbb{Z}$. For Tor , we take projective resolutions $F \rightarrow M$ and $P \rightarrow N$ of two arbitrary G -modules M and N and set

$$\text{Tor}_*^G(M, N) = H_*(F \otimes_G N) = H_*(F \otimes_G P) = H_*(M \otimes_G P).$$

We recover $H_*(G, —)$ as $\text{Tor}_*^G(\mathbb{Z}, —)$.

Similarly, if $F \rightarrow M$ is again a projective resolution, then we set

$$\text{Ext}_G^*(M, N) = H^*(\mathcal{H}om_G(F, N)).$$

We recover $H^*(G, —)$ as $\text{Ext}_G^*(\mathbb{Z}, —)$. The reader who has done exercise 4 of §1 will note that we could also take an injective resolution $N \rightarrow Q$ and write

$$\text{Ext}_G^*(M, N) = H^*(\mathcal{H}om_G(F, Q)) = H^*(\mathcal{H}om_G(M, Q)).$$

We do not intend to systematically develop the properties of Tor and Ext, but it will be useful to record a few results about them for future reference. The first shows that one has considerably more flexibility in the choice of resolutions for computing Tor than is apparent from the definition.

(2.1) Proposition. *Let $\varepsilon: F \rightarrow M$ and $\eta: P \rightarrow N$ be resolutions, not necessarily projective. If either F or P is a complex of flat modules, then $\text{Tor}_*^G(M, N) \approx H_*(F \otimes_G P)$.*

PROOF. Suppose, for instance, that F is flat, and let $\tilde{\eta}: \tilde{P} \rightarrow N$ be a projective resolution. By the fundamental lemma I.7.4, there is an augmentation-preserving chain map $f: \tilde{P} \rightarrow P$. Two applications of I.8.6 now yield

$$H_*(F \otimes_G P) \xleftarrow{\sim} H_*(F \otimes_G \tilde{P}) \xrightarrow{\sim} H_*(M \otimes_G \tilde{P}) = \text{Tor}_*^G(M, N),$$

where the first map is induced by f and the second by ε . □

One way to construct a resolution of a module M is to start with a resolution $\varepsilon: F \rightarrow \mathbb{Z}$ and tensor it with M to obtain $\varepsilon \otimes M: F \otimes M \rightarrow \mathbb{Z} \otimes M = M$. This will be a resolution under mild hypotheses on F and/or M . Suppose, for instance, that F is $\mathbb{Z}$ -free; then ε is a homotopy equivalence if we ignore the G -action (cf. I.7.6), so the same is true of $\varepsilon \otimes M$. [Alternatively, use the universal coefficient theorem to compute that

$$H_*(F \otimes M) = H_*(F) \otimes M = M.]$$

This method will be used now to show that Tor_*^G and Ext_G^* can often be computed in terms of $H_*(G, —)$ and $H^*(G, —)$.

(2.2) Proposition. *Let M and N be G -modules. If M is $\mathbb{Z}$ -torsion-free then*

$$\text{Tor}_*^G(M, N) \approx H_*(G, M \otimes N),$$

where G acts diagonally on $M \otimes N$. If M is $\mathbb{Z}$ -free, then

$$\text{Ext}_G^*(M, N) \approx H^*(G, \text{Hom}(M, N)),$$

where G acts diagonally on $\text{Hom}(M, N)$.

PROOF. Let $\varepsilon: F \rightarrow \mathbb{Z}$ be a projective resolution, and consider the resolution $\varepsilon \otimes M: F \otimes M \rightarrow M$. This is a flat resolution if M is $\mathbb{Z}$ -torsion-free and a

projective resolution if M is $\mathbb{Z}$ -free, by exercises 1 and 2 of §0. So if M is $\mathbb{Z}$ -torsion-free, then we have

$$\begin{aligned}\mathrm{Tor}_*^G(M, N) &\approx H_*((F \otimes M) \otimes_G N) && \text{by 2.1} \\ &= H_*((F \otimes M \otimes N)_G) \\ &= H_*(F \otimes_G (M \otimes N)) \\ &= H_*(G, M \otimes N).\end{aligned}$$

And if M is $\mathbb{Z}$ -free, then

$$\begin{aligned}\mathrm{Ext}_G^*(M, N) &= H^*(\mathcal{H}om_G(F \otimes M, N)) && \text{by definition of Ext} \\ &= H^*(\mathcal{H}om(F \otimes M, N)^G) \\ &= H^*(\mathcal{H}om(F, \mathrm{Hom}(M, N))^G) \\ &= H^*(\mathcal{H}om_G(F, \mathrm{Hom}(M, N))) \\ &= H^*(G, \mathrm{Hom}(M, N)).\end{aligned}$$

□

We will outline alternative proofs of this proposition in exercise 2 of §7 below.

Finally, we remark that Tor and Ext can be defined over an arbitrary ring R , using the same definitions as we gave for G -modules. One need only be careful about which modules are left modules and which are right ones. Thus $\mathrm{Tor}_*^R(M, N)$ is defined when M is a right module and N is a left module, whereas $\mathrm{Ext}_R^*(M, N)$ is defined when M and N are either both left modules or both right modules.

EXERCISE

Show that $\mathrm{Tor}_*^G(M, N) \approx \mathrm{Tor}_*^G(N, M)$. [Hint: See exercise 5 of §1.0.]

3 Extension and Co-extension of Scalars

Before proceeding further with the study of $H_*(G, M)$ and $H^*(G, M)$, we digress to discuss, in this and the next two sections, some module-theoretic constructions which play a fundamental role in the homology and cohomology theory of groups. Let $\alpha: R \rightarrow S$ be a ring homomorphism. Then any S -module can be regarded as an R -module via α , and we obtain in this way a functor from S -modules to R -modules, called *restriction of scalars*. The purpose of this section is to study two constructions which go in the opposite direction, from R -modules to S -modules.

For any (left) R -module M , consider the tensor product $S \otimes_R M$, where S is regarded as a right R -module by $s \cdot r = s\alpha(r)$. Since the natural left action of S on itself commutes with this right action of R on S , we can make $S \otimes_R M$ a (left) S -module by setting $s \cdot (s' \otimes m) = ss' \otimes m$. This S -module is said to be obtained from M by *extension of scalars* from R to S .

Note that there is a natural map $i: M \rightarrow S \otimes_R M$ given by $i(m) = 1 \otimes m$. Since $1 \otimes rm = \alpha(r) \otimes m = \alpha(r) \cdot (1 \otimes m)$ for $r \in R$, we have

$$(3.1) \quad i(rm) = \alpha(r)i(m);$$

in other words, i is an R -module map, where the S -module $S \otimes_R M$ is regarded as an R -module by restriction of scalars. Moreover, the following *universal mapping property* holds:

- (3.2) Given an S -module N and an R -module map $f: M \rightarrow N$, there is a unique S -module map $g: S \otimes_R M \rightarrow N$ such that $gi = f$:

$$\begin{array}{ccc} M & \xrightarrow{i} & S \otimes_R M \\ f \downarrow & \swarrow g & \\ N & & \end{array}$$

[Thus we have

$$(3.3) \quad \text{Hom}_S(S \otimes_R M, N) \approx \text{Hom}_R(M, N),$$

showing that the extension of scalars functor (R -modules) $\rightarrow$ (S -modules) is left adjoint to the restriction of scalars functor (S -modules) $\rightarrow$ (R -modules).] Heuristically, 3.2 says that $S \otimes_R M$ is the smallest S -module which receives an R -module map from M . To prove 3.2 we need only note that g , if it exists, must satisfy $g(s \otimes m) = sg(1 \otimes m) = sg(i(m)) = sf(m)$. This proves uniqueness and tells us how to define g in order to prove existence; the remaining details are left to the reader.

Note that 3.2 can be applied, in particular, with $M = N$ (regarded as an R -module by restriction of scalars) and $f = \text{id}_N$. We obtain, then, for any S -module N , a canonical S -module map

$$(3.4) \quad S \otimes_R N \rightarrow N,$$

given by $s \otimes n \mapsto sn$. This map is *surjective*; moreover, as an R -module map it is a *split surjection*.

We now consider a dual construction, which uses Hom instead of $\otimes$. Given a (left) R -module M , consider the abelian group $\text{Hom}_R(S, M)$, where S is regarded as a left R -module by $r \cdot s = \alpha(r)s$. Since the natural right action of S on itself commutes with this left action of R on S , we can make $\text{Hom}_R(S, M)$ a left S -module by setting $(sf)(s') = f(s's)$ for $f \in \text{Hom}_R(S, M)$. [Note: It is because of the contravariance of Hom in the first variable that the *right* action of S on itself induces a *left* action of S on $\text{Hom}_R(S, M)$.] This S -module is said to be obtained from M by *co-extension of scalars* from R to S .

There is a natural map $\pi: \text{Hom}_R(S, M) \rightarrow M$, given by $\pi(f) = f(1)$. Note that $\pi(\alpha(r)f) = (\alpha(r)f)(1) = f(\alpha(r)) = rf(1) = r\pi(f)$, so π is an

R -module map if the S -module $\text{Hom}_R(S, M)$ is regarded as an R -module by restriction of scalars. Moreover, we have:

- (3.5) Given an S -module N and an R -module map $f: N \rightarrow M$, there is a unique S -module map $g: N \rightarrow \text{Hom}_R(S, M)$ such that $\pi g = f$:

$$\begin{array}{ccc} & \text{Hom}_R(S, M) & \\ & \downarrow \pi & \\ N & \xrightarrow{f} & M \end{array}$$

(Note: A dashed arrow labeled g points from N to $\text{Hom}_R(S, M)$.)

[Thus

$$(3.6) \quad \text{Hom}_S(N, \text{Hom}_R(S, M)) \approx \text{Hom}_R(N, M),$$

so that co-extension of scalars is right adjoint to restriction of scalars.] Heuristically, 3.5 says that $\text{Hom}_R(S, M)$ is the smallest S -module which maps to M by an R -module map. To prove 3.5, note that g must satisfy $sg(n) = g(sn)$ for $s \in S$, $n \in N$; evaluating both sides at 1, we find $g(n)(s) = g(sn)(1) = \pi(g(sn)) = f(sn)$; existence and uniqueness of g follow easily.

Taking $M = N$ (regarded as an R -module) and $f = \text{id}_N$, we obtain from 3.5 a canonical S -module map

$$(3.7) \quad N \rightarrow \text{Hom}_R(S, N),$$

given by $n \mapsto (s \mapsto sn)$. This map is *injective*; moreover, as an R -module map it is a *split injection*.

Examples of extension and co-extension of scalars have already occurred in these notes. Namely, if α is the augmentation map $\varepsilon: \mathbb{Z}G \rightarrow \mathbb{Z}$, then the “extension of scalars” functor $(\mathbb{Z}G\text{-modules}) \rightarrow (\mathbb{Z}\text{-modules})$ is simply $M \mapsto M_G$, and the “co-extension of scalars” functor is $M \mapsto M^G$. More generally, exercise 3 of §II.2 treated the extension of scalars relative to a surjection $\mathbb{Z}G \rightarrow \mathbb{Z}[G/H]$, where $H \triangleleft G$. What we are primarily interested in, however, is the case where α is an inclusion of the form $\mathbb{Z}H \hookrightarrow \mathbb{Z}G$, where $H \subseteq G$. This case will be discussed in detail in §5 below.

EXERCISES

1. Show that extension of scalars takes projective R -modules to projective S -modules. [Hint: If P is a projective R -module, 3.3 shows that $\text{Hom}_S(S \otimes_R P, -)$ is an exact functor.]
2. Recall from exercise 4 of §I.7 that an R -module Q is called *injective* if the functor $\text{Hom}_R(-, Q)$ is exact. Show that co-extension of scalars takes injective R -modules to injective S -modules. [Hint: If Q is an injective R -module, 3.6 shows that $\text{Hom}_S(-, \text{Hom}_R(S, Q))$ is an exact functor.]
3. If S is flat as a right R -module (i.e., if $S \otimes_R -$ is an exact functor), show that restriction of scalars takes injective S -modules to injective R -modules. [Similar hint.]

4. If S is projective as a left R -module, show that restriction of scalars takes projective S -modules to projective R -modules. [Similar hint.]

The reader familiar with adjoint functors may wish to formulate a general statement of which 1–4 are special cases. [Note: Exercises 1 and 4 can also be done in a more concrete way, using the fact that a projective module is a direct summand of a free module. A special case of exercise 4 was already done in this way in exercise 2 of §1.8.]

4 Injective Modules

Recall from exercise 4 of §1.7 that an R -module Q is called *injective* if it satisfies the following equivalent conditions, dual to those by which we defined “projective”:

- (i) $\text{Hom}_R(-, Q)$ is exact.
- (ii) Every mapping problem

$$\begin{array}{ccccc}
 M' & \longrightarrow & M & \longrightarrow & M'' \\
 & \searrow \scriptstyle 0 & \downarrow & \swarrow \text{---} & \\
 & & Q & &
 \end{array}$$

with exact row can be solved.

- (iii) For any inclusion $M' \hookrightarrow M$, every map $M' \rightarrow Q$ can be extended to a map $M \rightarrow Q$:

$$\begin{array}{ccc}
 M' & \hookrightarrow & M \\
 \downarrow & \nearrow \text{---} & \\
 & & Q.
 \end{array}$$

The purpose of this section is to prove that every module can be embedded in an injective module; this is analogous to the (obvious) fact that every module is a quotient of a projective module.

(4.1) Proposition. *An R -module Q is injective if and only if every map $I \rightarrow Q$, where I is a left ideal of R , extends to a map $R \rightarrow Q$.*

PROOF. The “only if” part is trivial. To prove the “if” part, suppose that every map $I \rightarrow Q$ extends to R for all ideals $I \subset R$. It follows at once that if $C' \subset C$, where C is a cyclic R -module, then every map $C' \rightarrow Q$ extends to a map $C \rightarrow Q$. Now let $M' \subset M$ be an arbitrary inclusion. Given $f: M' \rightarrow Q$, there exists by Zorn’s lemma a maximal extension $F: M'' \rightarrow Q$ of f , where $M' \subseteq M'' \subseteq M$. If $M'' \neq M$ then there is a cyclic module $C \subseteq M$ such that $C \not\subseteq M''$.

But we know that $F|C \cap M''$ extends to C , so F extends to $M'' + C$, contradicting the maximality of F . Thus $M'' = M$ and Q is injective. $\square$

(4.2) Corollary. *If R is a principal ideal domain, then an R -module Q is injective if and only if it is divisible, i.e., if and only if $rQ = Q$ for every $r \neq 0$ in R .*

The proof is immediate.

As an example of the corollary, the $\mathbb{Z}$ -module Q is divisible and hence injective. Similarly, $Q/\mathbb{Z}$ is injective. The latter module is particularly interesting, because of the following result:

(4.3) Proposition. *If A is a $\mathbb{Z}$ -module and $0 \neq a \in A$, then there is a map $f: A \rightarrow Q/\mathbb{Z}$ such that $f(a) \neq 0$.*

PROOF. Since $Q/\mathbb{Z}$ contains $(1/n)\mathbb{Z}/\mathbb{Z} \approx \mathbb{Z}/n\mathbb{Z}$ for every $n \neq 0$, it is clear that there is a map $f_0: \mathbb{Z}a \rightarrow Q/\mathbb{Z}$ such that $f_0(a) \neq 0$. But $Q/\mathbb{Z}$ is injective, so f_0 extends to a map $f: A \rightarrow Q/\mathbb{Z}$. $\square$

This shows that $Q/\mathbb{Z}$ plays a role in abelian group theory dual to that played by $\mathbb{Z}$; for the latter admits non-zero maps to every non-zero abelian group.

Returning now to an arbitrary ring R , let R' be the R -module $\text{Hom}_R(R, Q/\mathbb{Z})$ obtained from $Q/\mathbb{Z}$ by co-extension of scalars with respect to the ring homomorphism $\mathbb{Z} \rightarrow R$. It is injective by exercise 2 of §3. As we will see, it plays the role in the theory of injectives that $R = R \otimes_{\mathbb{Z}} \mathbb{Z}$ plays in the theory of projectives. By a *co-free* module we mean an arbitrary direct product of copies of R' . The following result is the analogue of I.7.2:

(4.4) Lemma. *Co-free modules are injective.*

PROOF. It is immediate from the definition of "injective" that an arbitrary direct product of injectives is injective. $\square$

Recall now the standard proof that every module M is a quotient of a free module F ; one takes $F = \bigoplus R$, where there is one summand for each element of M , i.e., for each R -module map $R \rightarrow M$. Dually, we can consider $Q = \prod_{f \in \mathcal{F}} R'$, where $\mathcal{F} = \text{Hom}_R(M, R')$. There is an obvious map $i: M \rightarrow Q$, whose f -component for $f \in \mathcal{F}$ is the map f .

(4.5) Theorem. *The map i is a monomorphism. Consequently, any R -module can be embedded in an injective module.*

PROOF. If $0 \neq m \in M$, there is by 4.3 a $\mathbb{Z}$ -module map $f_0: M \rightarrow Q/\mathbb{Z}$ with $f_0(m) \neq 0$. By 3.5, f_0 factors through an R -module map $f: M \rightarrow R'$. It follows that $f(m) \neq 0$, and hence that $i(m) \neq 0$. $\square$

See Cartan–Eilenberg [1956], I.3.3, for a different proof of the second assertion of 4.5, based directly on 4.1.

EXERCISE

Let $R = \mathbb{Z}/n\mathbb{Z}$. Show that R is an injective R -module. Deduce:

- (a) If A is an abelian group such that $nA = 0$, and $C \subseteq A$ is a cyclic subgroup of order n , then C is a direct summand of A .
- (b) If A is as in (a), then A is a direct sum of cyclic groups.

5 Induced and Co-induced Modules

In this section we apply the constructions of §3 to ring homomorphisms of the form $\mathbb{Z}H \hookrightarrow \mathbb{Z}G$, where $H \subset G$. In this case extension of scalars (resp. co-extension of scalars) is called *induction* (resp. *co-induction*) from H to G . We will often write

$$\mathbb{Z}G \otimes_{\mathbb{Z}H} M = \text{Ind}_H^G M$$

and

$$\text{Hom}_{\mathbb{Z}H}(\mathbb{Z}G, M) = \text{Coind}_H^G M$$

for an H -module M . Note that a module of the form $\text{Ind}_{\{1\}}^G M$ is precisely what we called an *induced module* in exercise 1b of §1 above. Similarly, a module of the form $\text{Coind}_{\{1\}}^G M$ will be called *co-induced*.

Since the right translation action of H on G is free, $\mathbb{Z}G$ is a free right $\mathbb{Z}H$ -module; as basis we can take any set E of representatives for the left cosets gH . It follows that $\mathbb{Z}G \otimes_{\mathbb{Z}H} M$, as abelian group, admits a decomposition

$$\mathbb{Z}G \otimes_{\mathbb{Z}H} M = \bigoplus_{g \in E} g \otimes M,$$

where $g \otimes M = \{g \otimes m : m \in M\}$ and $g \otimes M \approx M$ via $g \otimes m \leftrightarrow m$. In particular, since we can take 1 as the representative of its coset, it follows that the canonical H -map $i: M \rightarrow \mathbb{Z}G \otimes_{\mathbb{Z}H} M$ defined in §3 maps M isomorphically onto its image $1 \otimes M$. We can therefore use i to regard M as an H -submodule of $\text{Ind}_H^G M$. Moreover, the summand $g \otimes M$ which occurs above is simply the transform of this submodule under the action of g , since $g \cdot (1 \otimes m) = g \otimes m$. We have therefore established:

(5.1) Proposition. *The G -module $\text{Ind}_H^G M$ contains M as an H -submodule and is the direct sum of the transforms gM , where g ranges over any set of representatives for the left cosets of H in G .*

More briefly, the second assertion of the proposition says:

$$(5.2) \quad \text{Ind}_H^G M = \bigoplus_{g \in G/H} gM.$$

This makes sense because M is mapped onto itself by the action of H , so that the subgroup gM of $\text{Ind}_H^G M$ depends only on the class of g in G/H .

The description 5.2 completely characterizes G -modules of the form $\text{Ind}_H^G M$. More precisely, suppose N is a G -module whose underlying abelian group is a direct sum $\bigoplus_{i \in I} M_i$. Assume that the G -action transitively permutes the summands, in the sense that there is a transitive action of G on I such that $gM_i = M_{gi}$ for all $g \in G$ and $i \in I$. Then we have:

(5.3) Proposition. *Let N be a G -module as above, let M be one of the summands M_i , and let $H \subseteq G$ be the isotropy group of i . Then M is an H -module and $N \approx \text{Ind}_H^G M$.*

PROOF. It is obvious that M is an H -submodule of N , and 3.2 implies that the inclusion $M \hookrightarrow N$ extends to a G -map $\text{Ind}_H^G M \rightarrow N$. Clearly φ maps the summand gM of $\text{Ind}_H^G M$ isomorphically onto the corresponding summand M_{gi} of N , so φ is an isomorphism. $\square$

(5.4) Corollary. *Let N be a G -module whose underlying abelian group is of the form $\bigoplus_{i \in I} M_i$. Assume that the G -action permutes the summands according to some action of G on I . Let G_i be the isotropy group of i and let E be a set of representatives for $I \bmod G$. Then M_i is a G_i -module and there is a G -isomorphism $N \approx \bigoplus_{i \in E} \text{Ind}_{G_i}^G M_i$.*

PROOF. We have $I = \bigsqcup_{i \in E} G_i$, so $N = \bigoplus_{i \in E} \bigoplus_{j \in G_i} M_j$; now apply 5.3 to the inner sum. $\square$

(5.5) EXAMPLES. (a) The permutation module $\mathbb{Z}[G/H]$ is isomorphic to $\text{Ind}_H^G \mathbb{Z}$, with H acting trivially on $\mathbb{Z}$. This can be seen directly from the definition of $\text{Ind}_H^G \mathbb{Z}$, or, alternatively, by writing $\mathbb{Z}[G/H]$ as a direct sum of copies of $\mathbb{Z}$ and applying 5.3.

(b) Let X be a G -CW-complex and consider the G -module $C_n(X)$. This is a direct sum of copies of $\mathbb{Z}$, one for each n -cell of X , and the summands are permuted by the G -action. Hence 5.4 gives

$$C_n(X) \approx \bigoplus_{\sigma \in \Sigma_n} \text{Ind}_{G_\sigma}^G \mathbb{Z}_\sigma,$$

where Σ_n is a set of representatives for the G -orbits of n -cells,

$$G_\sigma = \{g \in G : g\sigma = \sigma\},$$

and $\mathbb{Z}_\sigma$ is the “orientation module” associated to σ , i.e., $\mathbb{Z}_\sigma$ is an infinite cyclic group whose two generators correspond to the two orientations of σ . (Thus $g \in G_\sigma$ acts on $\mathbb{Z}_\sigma$ as $+1$ if g preserves the orientation of σ and -1 otherwise.) Note that if G acts freely on X , then the isomorphism above simply reduces to our observation in §1.4 that $C_n(X)$ is a free $\mathbb{Z}G$ -module with one basis element for each $\sigma \in \Sigma_n$.

Before stating our next result we remark that the summand gM of $\text{Ind}_H^G M$ is closed under the action of gHg^{-1} , hence gM is a gHg^{-1} -module. Note that the action of g^{-1} gives a bijection $f: gM \xrightarrow{\sim} M$ such that $f(kn) = g^{-1}kg \cdot f(n)$ for $k \in gHg^{-1}$, $n \in gM$. Hence gM can be identified with the gHg^{-1} -module obtained from the H -module M by “restriction of scalars” via the conjugation isomorphism $gHg^{-1} \xrightarrow{\sim} H$.

For any G -module N we denote by $\text{Res}_H^G N$ the H -module obtained by restriction of scalars from G to H .

(5.6) Proposition. (a) *Let N be a G -module. Then*

$$\text{Ind}_H^G \text{Res}_H^G N \approx \mathbb{Z}[G/H] \otimes N,$$

where G acts diagonally on the tensor product.

(b) *Let H and K be subgroups of G and let E be a set of representatives for the double cosets KgH . For any H -module M , there is a K -isomorphism*

$$\text{Res}_K^G \text{Ind}_H^G M \approx \bigoplus_{g \in E} \text{Ind}_{K \cap gHg^{-1}}^K \text{Res}_{K \cap gHg^{-1}}^{gHg^{-1}} gM.$$

In particular, if $H \triangleleft G$, then there is an H -isomorphism

$$\text{Res}_H^G \text{Ind}_H^G M \approx \bigoplus_{g \in G/H} gM.$$

Remark. In view of the observations preceding the statement of the proposition, we can identify the module $\text{Res}_{K \cap gHg^{-1}}^{gHg^{-1}} gM$ which occurs above with the module $\text{Res}_{K \cap gHg^{-1}}^H M$, where the latter restriction is with respect to the conjugation map $K \cap gHg^{-1} \rightarrow H$, $k \mapsto g^{-1}kg$.

PROOF OF 5.6. (a) will be left to the reader; it is the special case $M = \mathbb{Z}$ of exercise 2a below. For (b) we note that the summands of $\text{Ind}_H^G M = \bigoplus_{g \in G/H} gM$ are permuted by the action of K according to the natural action of K on G/H . Since the K -orbits in G/H correspond to double cosets KgH and the isotropy group in K of the coset gH is $K \cap gHg^{-1}$, (b) follows from 5.4. $\square$

We will have several occasions to use 5.6a in the special case $H = \{1\}$. The content of the result in this case is:

(5.7) Corollary. *Let M be a G -module and let M_0 be its underlying abelian group. Then $\mathbb{Z}G \otimes M$ (with diagonal G -action) is canonically isomorphic to the induced module $\mathbb{Z}G \otimes M_0$. In particular, $\mathbb{Z}G \otimes M$ is a free $\mathbb{Z}G$ -module if M is free as a $\mathbb{Z}$ -module.*

The interested reader can verify that the proof of 5.6a yields the specific isomorphism $\mathbb{Z}G \otimes M_0 \rightarrow \mathbb{Z}G \otimes M$ given by $g \otimes m \mapsto g \otimes gm$, with inverse $g \otimes m \mapsto g \otimes g^{-1}m$.

Co-induction has properties analogous to the properties of induction given above, but with $\bigoplus$ replaced by $\prod$. The precise statements, however, are somewhat more awkward and require some notation: If $\pi_1: N \twoheadrightarrow M_1$ and $\pi_2: N \twoheadrightarrow M_2$ are surjections of abelian groups, then we write $\pi_1 \sim \pi_2$ if there is an isomorphism $h: M_1 \xrightarrow{\sim} M_2$ such that $h\pi_1 = \pi_2$, or, equivalently, if $\ker \pi_1 = \ker \pi_2$. If $\pi: N \twoheadrightarrow M$ is a surjection and N has a G -module structure, then we denote by πg for $g \in G$ the surjection $N \twoheadrightarrow M$ defined by $(\pi g)(x) = \pi(gx)$. Finally, by a *direct product decomposition* of N we mean a family of surjections $(\pi_i: N \twoheadrightarrow M_i)_{i \in I}$ such that the corresponding map $N \rightarrow \prod_{i \in I} M_i$ is an isomorphism. It is now easy to state and prove analogues for co-induction of 5.1, 5.3, 5.4, and 5.6. For example, the analogue of 5.1 is that the underlying abelian group of $\text{Coind}_H^G M$ admits a direct product decomposition $(\pi g)_{g \in H \backslash G}$, where $\pi: \text{Coind}_H^G M \rightarrow M$ is the canonical H -module map defined in §3. (Note that $\pi h \sim \pi$ for $h \in H$, so the equivalence class of the surjection πg depends only on the class of g in $H \backslash G$.) Similarly, the analogue of 5.3 is:

(5.8) Proposition. *Let N be a G -module which, as an abelian group, admits a direct product decomposition $(\pi_i: N \twoheadrightarrow M_i)_{i \in I}$. Assume that there is a transitive right action of G on I such that $\pi_i g \sim \pi_{ig}$ for all $i \in I$ and $g \in G$. Let $\pi: N \twoheadrightarrow M$ be one of the π_i and let $H \subseteq G$ be the isotropy group of i . Then M inherits an H -module structure from N , and $N \approx \text{Coind}_H^G M$.*

The reader can similarly formulate the analogues of 5.4 and 5.6.

Finally, since a direct product indexed by a finite set can be identified with the corresponding direct sum, the following result is not surprising:

(5.9) Proposition. *If $(G:H) < \infty$ then $\text{Ind}_H^G M \approx \text{Coind}_H^G M$.*

PROOF. There is an H -map $\varphi_0: M \rightarrow \text{Hom}_H(\mathbb{Z}G, M)$ given by

$$\varphi_0(m)(g) = \begin{cases} gm & g \in H \\ 0 & g \notin H. \end{cases}$$

This extends to a G -map $\varphi: \mathbb{Z}G \otimes_{\mathbb{Z}H} M \rightarrow \text{Hom}_H(\mathbb{Z}G, M)$ by 3.2. Choosing coset representatives and expressing $\mathbb{Z}G \otimes_{\mathbb{Z}H} M$ (resp. $\text{Hom}_H(\mathbb{Z}G, M)$) as a direct sum (resp. direct product) of copies of M , one sees that φ can be identified with the canonical inclusion of the sum into the product. Consequently, φ is an isomorphism if $(G:H) < \infty$. Alternatively, one can set $\psi(f) = \sum_{g \in G/H} g \otimes f(g^{-1})$ and verify that $\psi = \varphi^{-1}$. $\square$

EXERCISES

1. Prove that induction is "invariant under conjugation," in the sense that $\text{Ind}_H^G M \approx \text{Ind}_{gHg^{-1}}^G {}^g M$ for any H -module M . Deduce that the K -module

$$\text{Ind}_{K \cap gHg^{-1}}^K \text{Res}_{K \cap gHg^{-1}}^{gHg^{-1}} M$$

which occurs in 5.6b depends (up to isomorphism) only on the class of g in $K \backslash G/H$.

2. (a) For any H -module M and G -module N prove that

$$N \otimes \text{Ind}_H^G M \approx \text{Ind}_H^G (\text{Res}_H^G N \otimes M),$$

where the tensor product on the left has the diagonal G -action and that on the right has the diagonal H -action. [Hint: Write the left side as $\bigoplus (N \otimes gM)$ and apply 5.3.]

(b) State and prove similar results for $\text{Hom}(\text{Ind}_H^G M, N)$ and $\text{Hom}(N, \text{Coind}_H^G M)$.

3. Use 5.7 to give a new proof of the result of exercise 2 of §0.

4. (a) If $(G:H) = \infty$, prove that $(\text{Ind}_H^G M)^G = 0$ for any H -module M .

(b) If G is finitely generated and $(G:H) = \infty$, prove that $(\text{Coind}_H^G M)_G = 0$ for any H -module M . [See Strebel [1977] for an algebraic proof of this. A topological proof can be given as follows. We may assume that G is a free group. In this case let Y be a finite, connected 1-complex with $\pi_1 Y = G$ and let $\tilde{Y}$ be the covering complex corresponding to H . Then M (resp. $\text{Coind}_H^G M$) can be regarded as a local coefficient system on $\tilde{Y}$ (resp. Y), and it is easy to see that $(\text{Coind}_H^G M)_G = H_0(Y, \text{Coind}_H^G M) = H_0^{(\infty)}(\tilde{Y}, M)$, where $H_^{(\infty)}$ denotes homology based on infinite chains. The result now follows from the following elementary fact: If X is an infinite, locally finite, connected complex, then $H_0^{(\infty)}(X, M) = 0$ for any local coefficient system M .]

*(c) Given $H \subseteq G$, prove that the following conditions are equivalent:

- (i) There is a finitely generated subgroup $G' \subseteq G$ such that $(G': G' \cap gHg^{-1}) = \infty$ for all $g \in G$. [If $H \triangleleft G$, for example, this just says that the group G/H contains a finitely generated infinite subgroup, i.e., G/H is not locally finite.]
- (ii) $(\text{Coind}_H^G M)_G = 0$ for all H -modules M .
- (iii) The element of $(\text{Coind}_H^G \mathbb{Z})_G$ represented by the augmentation map $\varepsilon \in \text{Hom}_H(\mathbb{Z}G, \mathbb{Z}) = \text{Coind}_H^G \mathbb{Z}$ is zero. [Hint: Use (b) and the double-coset formula.]

- *5. If G is finite and k is a field, show that a kG -module is projective iff it is injective. [Hint: Start by showing that the free module kG is injective; this follows from the analogue over k of 5.9, which implies that $\text{Hom}_{kG}(-, kG) \approx \text{Hom}_k(-, k)$.]

6 H_* and H^* as Functors of the Coefficient Module

Since $F \otimes_G -$ and $\mathcal{H}om_G(F, -)$ are covariant functors, it is clear that $H_*(G, -)$ and $H^*(G, -)$ are covariant functors of the coefficient module. The following proposition gives the basic properties of these functors.

(6.1) Proposition. (i) *There is a natural isomorphism $H_0(G, M) \approx M_G$.*

(i') *There is a natural isomorphism $H^0(G, M) \approx M^G$.*

(ii) *For any exact sequence $0 \rightarrow M' \xrightarrow{i} M \xrightarrow{j} M'' \rightarrow 0$ of G -modules and any integer n there is a natural map $\partial: H_n(G, M'') \rightarrow H_{n-1}(G, M')$ such that the sequence*

$$\cdots \rightarrow H_1(G, M) \rightarrow H_1(G, M'') \xrightarrow{\partial} H_0(G, M') \rightarrow H_0(G, M) \rightarrow H_0(G, M'') \rightarrow 0$$

is exact. (The unlabelled arrows here represent the maps induced by i and j .)

(ii') For any exact sequence as in (ii) and any integer n there is a natural map $\delta: H^n(G, M'') \rightarrow H^{n+1}(G, M')$ such that the sequence

$$0 \rightarrow H^0(G, M') \rightarrow H^0(G, M) \rightarrow H^0(G, M'') \xrightarrow{\delta} H^1(G, M') \rightarrow H^1(G, M) \rightarrow \dots$$

is exact.

(iii) If P is a projective $\mathbb{Z}G$ -module then $H_n(G, P) = 0$ for $n > 0$.

(iii') If Q is an injective $\mathbb{Z}G$ -module then $H^n(G, Q) = 0$ for $n > 0$.

Note: The naturality assertion in (ii) means that for any commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & M' & \longrightarrow & M & \longrightarrow & M'' \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & N' & \longrightarrow & N & \longrightarrow & N'' \longrightarrow 0 \end{array}$$

with exact rows, the square

$$\begin{array}{ccc} H_n(G, M'') & \xrightarrow{\delta} & H_{n-1}(G, M') \\ \downarrow & & \downarrow \\ H_n(G, N'') & \xrightarrow{\delta} & H_{n-1}(G, N') \end{array}$$

is commutative.

PROOF OF 6.1. (i) and (i') were proved in §1. Given an exact sequence as in (ii) and a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$, we have an exact sequence of chain complexes

$$0 \rightarrow M' \otimes_G F \rightarrow M \otimes_G F \rightarrow M'' \otimes_G F \rightarrow 0$$

since projectives are flat; the corresponding long exact homology sequence yields (ii). Similarly, (ii') follows from the sequence of cochain complexes

$$0 \rightarrow \mathcal{H}om_G(F, M') \rightarrow \mathcal{H}om_G(F, M) \rightarrow \mathcal{H}om_G(F, M'') \rightarrow 0,$$

which is exact by the definition of "projective." Finally, (iii) and (iii') are immediate consequences of the definitions of $H_*(G, -)$ and $H^*(G, -)$ and the exactness of $- \otimes_G P$ and $\mathcal{H}om_G(-, Q)$. $\square$

Properties (iii) and (iii') are usually expressed by saying that projective modules are H_* -acyclic and injective modules are H^* -acyclic.

A functor T (say from R -modules to abelian groups) is said to be *effaceable* if every module M is a quotient of a module $\bar{M}$ such that $T(\bar{M}) = 0$. Clearly T is effaceable if $T(P) = 0$ for every projective P . Conversely, if T is effaceable then $T(P) = 0$ for every projective P . For let $\pi: \bar{P} \rightarrow P$ be a surjection with $T(\bar{P}) = 0$; since P is projective, π must split, hence $T(\pi): T(\bar{P}) \rightarrow T(P)$ is a split surjection and $T(P) = 0$. Thus the content of (iii) above is

that $H_n(G, -)$ is effaceable for $n > 0$. Similarly, (iii') says that $H^n(G, -)$ is *co-effaceable* for $n > 0$, i.e., that every module M can be embedded in a module $\bar{M}$ such that $H^n(G, \bar{M}) = 0$.

(6.2) Proposition ("Shapiro's lemma"). *If $H \subseteq G$ and M is an H -module, then*

$$H_*(H, M) \approx H_*(G, \text{Ind}_H^G M)$$

and

$$H^*(H, M) \approx H^*(G, \text{Coind}_H^G M).$$

[Note: After we have discussed the functoriality of H_* and H^* with respect to group homomorphisms, we will be able to make 6.2 more precise by showing that the isomorphisms are induced by the inclusion $H \hookrightarrow G$ and the canonical H -maps $M \rightarrow \text{Ind}_H^G M$ and $\text{Coind}_H^G M \rightarrow M$. See exercise 2 of §8.]

PROOF. Let F be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$. Then F can also be regarded as a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}H$, so

$$H_*(H, M) \approx H_*(F \otimes_{\mathbb{Z}H} M).$$

But $F \otimes_{\mathbb{Z}H} M \approx F \otimes_{\mathbb{Z}G} (\mathbb{Z}G \otimes_{\mathbb{Z}H} M) \approx F \otimes_G (\text{Ind}_H^G M)$, whence the first isomorphism. The second isomorphism follows from the universal property of co-induction, which implies $\mathcal{H}om_H(F, M) \approx \mathcal{H}om_G(F, \text{Coind}_H^G M)$, cf. 3.6. $\square$

Taking $M = \mathbb{Z}$, for example, we conclude from 6.2 that

$$(6.3) \quad H_*(H) \approx H_*(G, \mathbb{Z}[G/H]).$$

(This shows that homology with coefficients is of interest even if one is primarily interested in ordinary integral homology). In case $(G:H) < \infty$, $\text{Coind}_H^G M \approx \text{Ind}_H^G M$ by 5.9, so we also have

$$(6.4) \quad H^*(H, \mathbb{Z}) \approx H^*(G, \mathbb{Z}[G/H]).$$

Similarly, still assuming $(G:H) < \infty$, we find

$$(6.5) \quad H^*(H, \mathbb{Z}H) \approx H^*(G, \mathbb{Z}G),$$

since $\mathbb{Z}G \otimes_{\mathbb{Z}H} \mathbb{Z}H \approx \mathbb{Z}G$. (It is also true, of course, that $H_*(H, \mathbb{Z}H) \approx H_*(G, \mathbb{Z}G)$, but this is of no interest in view of 6.1(iii).) Finally, we remark that Shapiro's lemma can be applied with $H = \{1\}$ to yield:

(6.6) Corollary. *Induced modules $\mathbb{Z}G \otimes A$ are H_* -acyclic. Co-induced modules $\text{Hom}(\mathbb{Z}G, A)$ are H^* -acyclic.*

Note that 6.6 yields another proof that $H_n(G, -)$ is effaceable and $H^n(G, -)$ is co-effaceable for $n > 0$; for we have seen (3.4 and 3.7) that every module is a quotient of an induced module and that every module can be embedded in a co-induced module.

EXERCISE

(a) Show that the Mayer-Vietoris sequence II.7.7 can be deduced from the short exact sequence II.7.8 of permutation modules. [Hint: Use 6.1(ii) and 6.3.]

(b) More generally, derive in this way a Mayer-Vietoris sequence for homology and cohomology with coefficients. [Apply $- \otimes M$ and $\text{Hom}(-, M)$ to II.7.8; use 5.6a and 6.2.]

7 Dimension Shifting

We saw in §6 that $H_n(G, -)$ is effaceable and $H^n(G, -)$ is co-effaceable for $n > 0$. The significance of this is that it allows us to use the following *dimension-shifting* technique: Given a module M , choose an H_* -acyclic module $\bar{M}$ which maps onto M (e.g., take $\bar{M} = \mathbb{Z}G \otimes M$), and let

$$K = \ker\{\bar{M} \rightarrow M\}.$$

Then the long exact sequence of 6.1(ii) yields

$$(7.1) \quad H_n(G, M) \approx \begin{cases} H_{n-1}(G, K) & n > 1 \\ \ker\{H_0(G, K) \rightarrow H_0(G, \bar{M})\} & n = 1. \end{cases}$$

Thus a question about H_n can, in principle, be reduced to a question about H_{n-1} , provided we are willing to change the coefficient module. Iterating this procedure, we are ultimately reduced to H_0 . Similarly, embedding M in an H^* -acyclic module $\bar{M}$ (e.g., $\bar{M} = \text{Hom}(\mathbb{Z}G, M)$) and letting

$$C = \text{coker}\{M \rightarrow \bar{M}\},$$

we find

$$(7.2) \quad H^n(G, M) \approx \begin{cases} H^{n-1}(G, C) & n > 1 \\ \text{coker}\{H^0(G, \bar{M}) \rightarrow H^0(G, C)\} & n = 1. \end{cases}$$

This argument shows, at least heuristically, that a “homology theory” H_* having properties analogous to (ii) and (iii) of 6.1 is completely determined by H_0 . Similarly, a “cohomology theory” H^* satisfying (ii') and (iii') is determined by H^0 . The rest of this section will be devoted to a precise formulation and proof of these assertions.

Let R be an arbitrary ring and $T = (T_n)_{n \in \mathbb{Z}}$ a family of covariant functors from R -modules to abelian groups. Assume that we are given “connecting homomorphisms” $\partial: T_n(M'') \rightarrow T_{n-1}(M')$ for every short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ of R -modules. We require that ∂ be *natural* (in the

sense explained after the statement of Proposition 5.1) and that all composites be zero in the sequence

$$(\dagger) \quad \cdots \rightarrow T_{n+1}(M'') \xrightarrow{\partial} T_n(M') \rightarrow T_n(M) \rightarrow T_n(M'') \xrightarrow{\partial} T_{n-1}(M') \rightarrow \cdots$$

We will then say that T (or, more precisely, (T, ∂)) is a ∂ -functor. If, in addition, $T_n = 0$ for $n < 0$ and $(\dagger)$ is exact for every short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$, then we will say that T is a *homological functor*. Thus, for example, the content of 6.1(ii) is that $H_*(G, -)$ is a homological functor on the category of G -modules.

If S and T are ∂ -functors, then a *map* from S to T is a family φ of natural transformations $\varphi_n: S_n \rightarrow T_n$ such that the square

$$\begin{array}{ccc} S_n(M'') & \xrightarrow{\partial} & S_{n-1}(M') \\ \varphi_n(M'') \downarrow & & \downarrow \varphi_{n-1}(M') \\ T_n(M'') & \xrightarrow{\partial} & T_{n-1}(M') \end{array}$$

commutes for every short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ and every n .

(7.3) Theorem. *Let H be a homological functor such that H_n is effaceable for $n > 0$. If T is an arbitrary ∂ -functor and $\varphi_0: T_0 \rightarrow H_0$ is a natural transformation, then φ_0 extends uniquely to a map $\varphi: T \rightarrow H$ of ∂ -functors. This map φ is an isomorphism if and only if the following three conditions hold:*

- (i) φ_0 is an isomorphism.
- (ii) T is homological.
- (iii) T_n is effaceable for $n > 0$.

Thus a homological functor H which is effaceable in positive dimensions is uniquely determined (up to canonical isomorphism of ∂ -functors) by H_0 . In particular, it follows that $H_*(G, -)$ is uniquely determined by the properties (i)–(iii) of 6.1.

PROOF OF 7.3. We wish to construct $\varphi_n: T_n \rightarrow H_n$ so that

$$(*) \quad \begin{array}{ccc} T_n(M'') & \xrightarrow{\partial} & T_{n-1}(M') \\ \varphi_n(M'') \downarrow & & \downarrow \varphi_{n-1}(M') \\ H_n(M'') & \xrightarrow{\partial} & H_{n-1}(M') \end{array}$$

commutes for every short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$. For $n \leq 0$ there is no choice and all diagrams $(*)$ commute automatically. Assume, then, that $n > 0$ and that φ_{n-1} has been defined. For any module M choose a

short exact sequence $0 \rightarrow K \rightarrow P \rightarrow M \rightarrow 0$ with P projective, and consider the diagram

$$\begin{array}{ccccc}
 T_n(M) & \xrightarrow{\partial} & T_{n-1}(K) & \longrightarrow & T_{n-1}(P) \\
 \downarrow \varphi_n(M) & & \downarrow \varphi_{n-1}(K) & & \downarrow \varphi_{n-1}(P) \\
 0 \longrightarrow & H_n(M) & \xrightarrow{\partial} & H_{n-1}(K) & \longrightarrow H_{n-1}(P)
 \end{array}
 \quad (**)$$

where the dotted arrow represents the map we want to define. Note that the composite on the top row is zero, that the bottom row is exact (because $H_n(P) = 0$), and that the right-hand square commutes (by naturality of φ_{n-1}). Consequently, there is a unique map $\varphi_n(M)$ which makes $(**)$ commute. Since this definition of φ_n was forced on us by $(*)$, the uniqueness assertion of the theorem is clear. It is also clear from $(**)$ that if conditions (i)–(iii) hold then we can prove inductively that φ_n is an isomorphism. It remains, therefore, to complete the proof of the existence of φ_n by proving that φ_n is well-defined and natural and that all squares of the form $(*)$ commute. We will need the following lemma.

(7.4) Lemma. *Let $\varphi_n(M)$ be defined as above by means of a short exact sequence $0 \rightarrow K \rightarrow P \rightarrow M \rightarrow 0$. For any diagram*

$$\begin{array}{ccccccc}
 & & & & M & & \\
 & & & & \downarrow f & & \\
 0 & \longrightarrow & N' & \longrightarrow & N & \longrightarrow & N'' \longrightarrow 0
 \end{array}$$

with exact row, the diagram

$$\begin{array}{ccccccc}
 T_n(M) & \xrightarrow{T_n(f)} & T_n(N'') & \xrightarrow{\partial} & T_{n-1}(N') & & \\
 \downarrow \varphi_n(M) & & & & \downarrow \varphi_{n-1}(N') & & \\
 H_n(M) & \xrightarrow{H_n(f)} & H_n(N'') & \xrightarrow{\partial} & H_{n-1}(N') & &
 \end{array}$$

commutes.

PROOF. Since P is projective, the given diagram can be completed to a commutative diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & K & \longrightarrow & P & \longrightarrow & M \longrightarrow 0 \\
 & & \downarrow g & & \downarrow & & \downarrow f \\
 0 & \longrightarrow & N' & \longrightarrow & N & \longrightarrow & N'' \longrightarrow 0.
 \end{array}$$

Consider now the diagram

$$\begin{array}{ccccc}
 T_n(M) & \xrightarrow{\quad \partial \quad} & T_{n-1}(K) & & \\
 \downarrow T_n(f) & \searrow \varphi_n(M) & \textcircled{1} & \swarrow \varphi_{n-1}(K) & \downarrow T_{n-1}(g) \\
 & H_n(M) & \xrightarrow{\quad \partial \quad} & H_{n-1}(K) & \\
 & \downarrow H_n(f) & \textcircled{2} & \downarrow H_{n-1}(g) & \textcircled{3} \\
 & H_n(N'') & \xrightarrow{\quad \partial \quad} & H_{n-1}(N') & \\
 & \swarrow \varphi_{n-1}(N') & & & \\
 T_n(N'') & \xrightarrow{\quad \partial \quad} & T_{n-1}(N') & &
 \end{array}$$

where the maps labelled ∂ in ① are the “ ∂ -maps” associated to $0 \rightarrow K \rightarrow P \rightarrow M \rightarrow 0$. Then ① commutes by (**), ② commutes because H is a ∂ -functor, and ③ commutes by naturality of φ_{n-1} . Moreover, the outer square commutes because T is a ∂ -functor. The lemma follows at once. $\square$

Returning now to the proof of the theorem, let $f: M_1 \rightarrow M_2$ be an arbitrary map of R -modules. Choose exact sequences $0 \rightarrow K_i \rightarrow P_i \rightarrow M_i \rightarrow 0$ with P_i projective ($i = 1, 2$), and use them to define $\varphi_n(M_i): T_n(M_i) \rightarrow H_n(M_i)$ as in (**). Applying the lemma to the diagram

$$\begin{array}{c}
 M_1 \\
 \downarrow f \\
 0 \longrightarrow K_2 \longrightarrow P_2 \longrightarrow M_2 \longrightarrow 0,
 \end{array}$$

we conclude that the outer rectangle is commutative in the diagram

$$\begin{array}{ccccc}
 T_n(M_1) & \xrightarrow{T_n(f)} & T_n(M_2) & \xrightarrow{\quad \partial \quad} & T_{n-1}(K_2) \\
 \downarrow \varphi_n(M_1) & & \downarrow \varphi_n(M_2) & & \downarrow \varphi_{n-1}(K_2) \\
 H_n(M_1) & \xrightarrow{H_n(f)} & H_n(M_2) & \xrightarrow{\quad \partial \quad} & H_{n-1}(K_2).
 \end{array}$$

Now the right-hand square is commutative by (**), and

$$\partial: H_n(M_2) \rightarrow H_{n-1}(K_2)$$

is injective; hence the left-hand square commutes. This shows that φ_n is natural. It also shows that $\varphi_n(M)$ is well-defined, independent of the choice of exact sequence $0 \rightarrow K \rightarrow P \rightarrow M \rightarrow 0$. [Take $f = \text{id}_M$.] Finally,

if $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is an arbitrary exact sequence, then we can apply the lemma to the diagram

$$\begin{array}{ccccccc} & & & & M'' & & \\ & & & & \parallel & & \\ 0 & \longrightarrow & M' & \longrightarrow & M & \longrightarrow & M'' \longrightarrow 0 \end{array}$$

to conclude that $(*)$ commutes. $\square$

Similarly one can define the notations of δ -functor and cohomological functor in the obvious way, and one can prove:

(7.5) Theorem. *Let H be a cohomological functor such that H^n is co-effaceable for $n > 0$. If T is an arbitrary δ -functor and $\varphi^0: H^0 \rightarrow T^0$ is a natural transformation, then φ^0 extends uniquely to a map $\varphi: H \rightarrow T$ of δ -functors. This map φ is an isomorphism if and only if the following three conditions hold:*

- (i) φ^0 is an isomorphism.
- (ii) T is cohomological.
- (iii) T^n is co-effaceable for $n > 0$.

EXERCISES

1. (a) Prove the following generalization of 1.4: Let $\cdots \rightarrow C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\varepsilon} M \rightarrow 0$ be exact, where each C_i is H_* -acyclic; then $H_*(G, M) \approx H_*(C_0)$. [Hint: Apply the dimension-shifting argument using the exact sequences $0 \rightarrow \ker \varepsilon \rightarrow C_0 \rightarrow M \rightarrow 0$, $0 \rightarrow \ker \partial \rightarrow C_1 \rightarrow \ker \varepsilon \rightarrow 0$, etc.]
 (b) Similarly, let $0 \rightarrow M \rightarrow C^0 \rightarrow C^1 \rightarrow \cdots$ be exact, where each C^i is H^* -acyclic. Prove that $H^*(G, M) \approx H^*(C^0)$.
2. Use the results of this section to give a new proof of Proposition 2.2. [Method 1: If M is $\mathbb{Z}$ -torsion-free, then $\text{Tor}_*^G(M, -)$ and $H_*(G, M \otimes -)$ are homological functors which agree in dimension 0 and are effaceable in positive dimensions. A similar argument applies to $\text{Ext}_G^*(M, -)$ and $H^*(G, \text{Hom}(M, -))$. Method 2: Compute $\text{Tor}_*^G(M, N)$ in terms of a projective resolution of M . Express this computation in the form $\text{Tor}_*^G(M, N) = H_*(C_G)$ and apply exercise 1a. Similarly, deduce the result on Ext from 1b.]
3. In the discussion at the beginning of this section, show that the surjection $\overline{M} \twoheadrightarrow M$ and the injection $M \hookrightarrow \overline{M}$ can always be taken to be $\mathbb{Z}$ -split [Hint: Use 3.4 and 3.7.]

8 H_* and H^* as Functors of Two Variables

Let $\mathcal{C}$ be the following category: An object of $\mathcal{C}$ is a pair (G, M) , where G is a group and M is a G -module; a map in $\mathcal{C}$ from (G, M) to (G', M') is a pair (α, f) , where $\alpha: G \rightarrow G'$ is a map of groups and $f: M \rightarrow M'$ is a map of abelian

groups such that $f(gm) = \alpha(g)f(m)$ for $g \in G, m \in M$. (In other words, f is a G -module homomorphism if M' is regarded as a G -module via α .) Given (α, f) , let F (resp. F') be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ (resp. $\mathbb{Z}G'$), and let $\tau: F \rightarrow F'$ be a chain map compatible with α as in §II.6. Then there is a chain map $\tau \otimes f: F \otimes_G M \rightarrow F' \otimes_{G'} M'$, and $\tau \otimes f$ induces a well-defined map $(\alpha, f)_*: H_*(G, M) \rightarrow H_*(G', M')$. In this way H_* becomes a covariant functor on $\mathcal{C}$. In case $M = M'$ and $f = \text{id}_{M'}$, we will simply write α_* for $(\alpha, f)_*: H_*(G, M') \rightarrow H_*(G', M')$. Note that the general map $(\alpha, f)_*$ can always be written as the composite

$$H_*(G, M) \xrightarrow{H_*(G, f)} H_*(G, M') \xrightarrow{\alpha_*} H_*(G', M'),$$

where $H_*(G, f)$ makes sense because f is a map of G -modules.

As an example of this functoriality we consider the maps induced by conjugation. Given $H \subseteq G$, a G -module M , and an element $g \in G$, we denote by $c(g): (H, M) \rightarrow (gHg^{-1}, M)$ the isomorphism in $\mathcal{C}$ given by

$$(h \mapsto ghg^{-1}, m \mapsto gm).$$

To compute $c(g)_*: H_*(H, M) \rightarrow H_*(gHg^{-1}, M)$ on the chain level, we choose a projective resolution F of $\mathbb{Z}$ over $\mathbb{Z}G$ and use F to compute the homology of H and gHg^{-1} . We can take $\tau: F \rightarrow F$ to be multiplication by g as in the proof of II.6.2, and we then find that $c(g)_*$ is induced by the chain map $F \otimes_H M \rightarrow F \otimes_{gHg^{-1}} M$ given by $x \otimes m \mapsto gx \otimes gm$. Note that this is just the map obtained from the usual diagonal action of g on $F \otimes M$ by passage to quotients.

For $z \in H_*(H, M)$ we set $gz = c(g)_* z \in H_*(gHg^{-1}, M)$. In view of the above description of $c(g)_*$ on the chain level, we have:

(8.1) Proposition. *If $h \in H$ then $hz = z$ for all $z \in H_*(H, M)$.*

(8.2) Corollary. *If $H \triangleleft G$ and M is a G -module, then the conjugation action of G on (H, M) induces an action of G/H on $H_*(H, M)$.*

The situation for cohomology is similar. Let $\mathcal{D}$ be the category with the same objects as $\mathcal{C}$, but where a map $(G, M) \rightarrow (G', M')$ is now a pair

$$(\alpha: G \rightarrow G', f: M' \rightarrow M);$$

as before we require that f be a G -module map, i.e., that $f(\alpha(g)m') = gf(m')$ for $g \in G, m' \in M'$. If F and F' are resolutions for G and G' and $\tau: F \rightarrow F'$ is a chain map compatible with α , then there is a chain map

$$\mathcal{H}om(\tau, f): \mathcal{H}om_G(F', M') \rightarrow \mathcal{H}om_G(F, M),$$

which induces $(\alpha, f)^*: H^*(G', M') \rightarrow H^*(G, M)$. Thus H^* is a contravariant functor on $\mathcal{D}$. In case $M = M'$ and $f = \text{id}_{M'}$, we will write α^* for $(\alpha, f)^*$; note that the general map $(\alpha, f)^*$ is the composite

$$H^*(G', M') \xrightarrow{\alpha^*} H^*(G, M') \xrightarrow{H^*(G, f)} H^*(G, M).$$

Let $\mathcal{C}_0$ be the subcategory of $\mathcal{C}$ consisting of all the objects (G, M) and of those maps (α, f) such that f is bijective. Then $\mathcal{C}_0$ is isomorphic to a subcategory of $\mathcal{D}$ by $(\alpha, f) \mapsto (\alpha, f^{-1})$, so H^* can be regarded as a contravariant functor on $\mathcal{C}_0$. In particular, the conjugation maps $c(g)$ discussed above are in $\mathcal{C}_0$, so we have an isomorphism $c(g)^*: H^*(gHg^{-1}, M) \rightarrow H^*(H, M)$ for any $g \in G$, $H \subseteq G$, and G -module M . If $z \in H^*(H, M)$ then we set

$$gz = (c(g)^*)^{-1}(z) \in H^*(gHg^{-1}, M).$$

In terms of a projective resolution F of $\mathbb{Z}$ over $\mathbb{Z}G$, the map $z \mapsto gz$ is induced by the cochain map $\mathcal{H}om_H(F, M) \rightarrow \mathcal{H}om_{gHg^{-1}}(F, M)$ given by

$$f \mapsto [x \mapsto gf(g^{-1}x)].$$

Note that this is just the usual diagonal action of g on $\mathcal{H}om(F, M)$, restricted to the subcomplex $\mathcal{H}om_H(F, M)$. In case $g \in H$ we clearly obtain the identity map, hence:

(8.3) Proposition. *If $h \in H$ then $hz = z$ for all $z \in H^*(H, M)$.*

(8.4) Corollary. *If $H \triangleleft G$ and M is a G -module, then the conjugation action of G on (H, M) induces an action of G/H on $H^*(H, M)$.*

EXERCISES

1. If H is central in G and M is an abelian group with trivial G -action, show that G/H acts trivially on $H_*(H, M)$ and $H^*(H, M)$.
2. Let $\alpha: H \hookrightarrow G$ be an inclusion, let M be an H -module, and let $i: M \rightarrow \text{Ind}_H^G M$ and $\pi: \text{Coind}_H^G M \rightarrow M$ be the canonical H -maps. Show that the isomorphisms of Shapiro's lemma (6.2) are given by $(\alpha, i)_*: H_*(H, M) \rightarrow H_*(G, \text{Ind}_H^G M)$ and $(\alpha, \pi)^*: H^*(G, \text{Coind}_H^G M) \rightarrow H^*(H, M)$. [Hint: You can take τ to be the identity map.]

9 The Transfer Map

Given an inclusion $\alpha: H \hookrightarrow G$ and a G -module M , we have seen that there are maps $\alpha^*: H^*(G, M) \rightarrow H^*(H, M)$ and $\alpha_*: H_*(H, M) \rightarrow H_*(G, M)$. We will often write $\alpha^* = \text{res}_H^G$ and call it a *restriction map*. Similarly, we write $\alpha_* = \text{cor}_H^G$ and call it a *corestriction map*. The purpose of this section is to show that if $(G:H) < \infty$ then there are maps going in the other direction, called *transfer maps*. This extremely useful observation is due to Eckmann [1953] and Artin–Tate [unpublished]. The existence of the transfer maps is somewhat more subtle than that of α_* and α^* ; in particular, they are *not* induced by maps in the categories $\mathcal{C}$ and $\mathcal{D}$ of §8. [The name “transfer map” comes from the fact that on H_1 it coincides with the transfer map of classical group theory, due to Schur [1902]; see exercise 2 below.]

We will explain the transfer maps from five different points of view, all of which are useful:

(A) For any G -module M and any $H \subseteq G$ we have a canonical surjection of G -modules

$$(9.1) \quad \mathbb{Z}G \otimes_{\mathbb{Z}H} M \rightarrow M$$

and a canonical injection

$$(9.2) \quad M \rightarrow \text{Hom}_{\mathbb{Z}H}(\mathbb{Z}G, M),$$

cf. 3.4 and 3.7. Applying $H_*(G, —)$ to 9.1 and using Shapiro's lemma, we obtain a map $H_*(H, M) \rightarrow H_*(G, M)$; this is easily seen to coincide with α_* . [Use exercise 2 of §8.] Similarly, applying $H^*(G, —)$ to 9.2 and using Shapiro's lemma we obtain $\alpha^*: H^*(G, M) \rightarrow H^*(H, M)$. If we assume now that $(G:H) < \infty$, then $\mathbb{Z}G \otimes_{\mathbb{Z}H} M \approx \text{Hom}_{\mathbb{Z}H}(\mathbb{Z}G, M)$ by 5.9. Consequently, we can apply $H^*(G, —)$ to 9.1 and $H_*(G, —)$ to 9.2 to obtain the transfer maps going in the other direction. These maps are often denoted

$$\text{cor}_H^G: H^*(H, M) \rightarrow H^*(G, M)$$

and

$$\text{res}_H^G: H_*(G, M) \rightarrow H_*(H, M)$$

or simply tr_H^G if it is clear from the context whether we are talking about H_* or H^* .

(B) For any $H \subseteq G$ we can regard both $H_*(G, —)$ and $H_*(H, —)$ as homological functors on the category of G -modules, and both are easily seen to be effaceable in positive dimensions. Assuming now that $(G:H) < \infty$, we define $\text{tr}: M_G \rightarrow M_H$ by $\text{tr}(\bar{m}) = \sum_{g \in H \backslash G} \bar{g}m$, where $\bar{m}$ (resp. $\bar{gm}$) denotes the image of m in M_G (resp. M_H). (The sum makes sense because $\bar{gm}$ depends only on the class of g in $H \backslash G$.) We now define $\text{res}: H_*(G, —) \rightarrow H_*(H, —)$ to be the unique extension of tr to a map of homological functors, cf. Theorem 7.3. To see that this agrees with the map defined in (A), one need only verify that the latter is compatible with ∂ and equals tr in dimension zero. Details are left to the reader. Similarly, $\text{cor}: H^*(H, —) \rightarrow H^*(G, —)$ can be defined as the unique map of cohomological functors which in dimension zero is the map $\text{tr}: M^H \rightarrow M^G$ defined by $\text{tr}(m) = \sum_{g \in G/H} gm$. [To verify co-effaceability of $H^n(H, —)$ on the category of G -modules, use exercise 3 of §3; alternatively, note that a co-induced G -module is also co-induced as an H -module.]

(C) Let F be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$. Then $F \otimes_H M = (F \otimes M)_H$ and $F \otimes_G M = (F \otimes M)_G$. We can now define $\text{res}: H_*(G, M) \rightarrow H_*(H, M)$ to be the map induced by the chain map $\text{tr}: (F \otimes M)_G \rightarrow (F \otimes M)_H$, where tr is as in (B). This defines a map of homological functors which equals

tr on H_0 , hence it agrees with definition (B). Similarly, $\mathcal{H}om_G(F, M) = \mathcal{H}om(F, M)^G$ and $\mathcal{H}om_H(F, M) = \mathcal{H}om(F, M)^H$, where G acts diagonally on $\mathcal{H}om$ (cf. §0). Hence there is a cochain map $\text{tr}: \mathcal{H}om(F, M)^H \rightarrow \mathcal{H}om(F, M)^G$ which induces $\text{cor}: H^*(H, M) \rightarrow H^*(G, M)$.

(D) If we wish to compute the transfer maps of (C) using one resolution $F(G)$ for G and a different one $F(H)$ for H , then we need an (augmentation-preserving) H -chain map $\tau: F(G) \rightarrow F(H)$ in order to realize the canonical isomorphism $H_*(F(G) \otimes_H M) \xrightarrow{\sim} H_*(F(H) \otimes_H M)$. The transfer maps will then be induced by the composites

$$(9.3) \quad F(G) \otimes_G M \xrightarrow{\text{tr}} F(G) \otimes_H M \xrightarrow{\tau \otimes M} F(H) \otimes_H M$$

and

$$(9.4) \quad \mathcal{H}om_H(F(H), M) \xrightarrow{\mathcal{H}om(\tau, M)} \mathcal{H}om_H(F(G), M) \xrightarrow{\text{tr}} \mathcal{H}om_G(F(G), M).$$

Taking $F(G)$ and $F(H)$ to be the standard resolutions, for example, we can easily write down such a map τ as follows. Choose a set of representatives for the right cosets Hg . Then there is a unique map $\rho: G \rightarrow H$ of left H -sets which sends every coset representative to 1; explicitly, if $\bar{g}$ denotes the representative of Hg , then $\rho g = g\bar{g}^{-1}$. Using the "homogeneous" description of the standard resolutions, we can now define $\tau: F(G) \rightarrow F(H)$ by $\tau(g_0, \dots, g_n) = (\rho g_0, \dots, \rho g_n)$. The interested reader can translate this into the bar notation and can write out the corresponding transfer maps $C_*(G, M) \rightarrow C_*(H, M)$ and $C^*(H, M) \rightarrow C^*(G, M)$ given by 9.3 and 9.4.

(E) If $\tilde{Y} \rightarrow Y$ is a covering map of finite degree, then there are transfer maps $H_*(Y, M) \rightarrow H_*(\tilde{Y}, M)$ and $H^*(\tilde{Y}, M) \rightarrow H^*(Y, M)$ for any coefficient system M on Y . In terms of cellular chain complexes, the homology transfer is induced by $\sigma \otimes m \mapsto \sum_{\bar{\sigma}} \bar{\sigma} \otimes m$, where σ is an oriented cell of Y and $\bar{\sigma}$ ranges over the oriented cells of $\tilde{Y}$ lying over σ . Similarly, the cohomology transfer is induced by the cochain map $f \mapsto [\sigma \mapsto \sum_{\bar{\sigma}} f(\bar{\sigma})]$. If Y is a $K(G, 1)$ and $\tilde{Y}$ is the covering space corresponding to $H \subseteq G$, then $\tilde{Y}$ is a $K(H, 1)$ and these transfer maps agree with the transfer maps $H_*(G, M) \rightarrow H_*(H, M)$ and $H^*(H, M) \rightarrow H^*(G, M)$. To see this, one need only apply definition (C) with F equal to the chain complex of the universal cover of Y .

We now give some properties of the transfer maps. Let $H(_, _)$ denote either H_* or H^* .

(9.5) Proposition. (i) Given $K \subseteq H \subseteq G$ with $(G: K) < \infty$,

$$\text{cor}_K^G = \text{cor}_H^G \circ \text{cor}_K^H \quad \text{and} \quad \text{res}_K^G = \text{res}_K^H \circ \text{res}_H^G.$$

(ii) Given $(G: H) < \infty$ and $z \in H(G, M)$, $\text{cor}_H^G \text{res}_H^G z = (G: H)z$.

(iii) Given $H, K \subseteq G$ with $(G: H) < \infty$ if $H(_, _) = H^*$ and $(G: K) < \infty$ if $H(_, _) = H_*$, we have

$$\text{res}_K^G \text{cor}_H^G z = \sum_{g \in E} \text{cor}_{K \cap gHg^{-1}}^K \text{res}_{K \cap gHg^{-1}}^{gHg^{-1}} gz$$

for any $z \in H(H, M)$, where E is a set of representatives for the double cosets KgH . In particular, if $H \triangleleft G$ and $(G:H) < \infty$, then $\text{res}_H^G \text{cor}_H^G z = \sum_{g \in G/H} gz = Nz$, where N is the norm element of $\mathbb{Z}[G/H]$.

(See §8 for the definition of gz . Note that in the cohomology case we can write $\text{res}_{K \cap gHg^{-1}}^{gHg^{-1}} gz = \text{res}_{K \cap gHg^{-1}}^H z$, where the “restriction map” on the right is with respect to the conjugation map ($k \mapsto g^{-1}kg$, $m \mapsto g^{-1}m$), regarded as a map $(K \cap gHg^{-1}, M) \rightarrow (H, M)$ in $\mathcal{C}_0$.)

PROOF OF 9.5: (i) and (ii) follow easily from the definitions and will be left to the reader. (iii) can be deduced from the double coset formula 5.6b and definition (A) of the transfer, but it is easier to work directly with definition (C). We will give the proof in the case $H(_, _) = H^*$; the homology case is similar. Let $u \in \mathcal{H}om(F, M)^H$ represent z . Then $\text{res}_K^G \text{cor}_H^G z$ is represented by $\sum_{g \in G/H} gu \in \mathcal{H}om(F, M)^G \subseteq \mathcal{H}om(F, M)^K$. Grouping together those terms gu corresponding to a given K -orbit in G/H , we find

$$\sum_{g \in G/H} gu = \sum_{g \in E} \sum_{k \in K/K_g} kg u,$$

where $K_g = \{k \in K: kgH = gH\} = K \cap gHg^{-1}$. Since the inner sum on the right clearly represents $\text{cor}_{K_g}^K \text{res}_{K_g}^{gHg^{-1}} gz$, this proves (iii). $\square$

EXERCISES

1. Show that $\text{res}_H^G: H(G, M) \rightarrow H(H, M)$ and $\text{cor}_H^G: H(H, M) \rightarrow H(G, M)$ are “invariant under conjugation,” in the sense that $g \cdot \text{res}_H^G w = \text{res}_{gHg^{-1}}^G w$ for $w \in H(G, M)$ and $\text{cor}_{gHg^{-1}}^G gz = \text{cor}_H^G z$ for $z \in H(H, M)$.
2. The transfer map $H_1(G) \rightarrow H_1(H)$ can be regarded as a map $G_{ab} \rightarrow H_{ab}$. Using definition (D), show that this map is given by

$$g \bmod [G, G] \mapsto \prod_{g' \in E} g' g \overline{g'}^{-1} \bmod [H, H],$$

where E is a set of representatives for the right cosets Hg and $\bar{x}$ is the representative of Hx . This formula is usually taken as the definition of the transfer map in group theory texts, cf. Hall [1959], formula (14.2.4).

10 Applications of the Transfer

To simplify the notation we will state the results of this section only for cohomology; similar results hold for homology.

(10.1) Proposition. *Let M be a G -module and $H \subseteq G$ a subgroup of finite index such that $H^n(H, M) = 0$ for some n . Then $H^n(G, M)$ is annihilated by $(G:H)$. In particular, if $(G:H)$ is invertible in M (i.e., if multiplication by $(G:H)$ is an isomorphism), then $H^n(G, M) = 0$.*

PROOF. The first assertion follows immediately from 9.5(ii). The second follows from the first, since $(G: H)$ is invertible in $H^n(G, M)$ if it is invertible in M . $\square$

Note that 10.1 applies to any $n > 0$ if G is finite and $H = \{1\}$. Hence:

(10.2) Corollary. *If G is finite, then $H^n(G, M)$ is annihilated by $|G|$ for all $n > 0$. If $|G|$ is invertible in M (e.g., if M is a $\mathbb{Q}G$ -module), then $H^n(G, M) = 0$ for all $n > 0$.*

It follows that $H^n(G, M)$ admits a primary decomposition

$$H^n(G, M) = \bigoplus_p H^n(G, M)_{(p)},$$

where p ranges over the primes dividing $|G|$ and $H^n(G, M)_{(p)}$ is the p -primary component of $H^n(G, M)$. Now fix a prime p and let H be a p -Sylow subgroup of G . Since $(G: H)$ is relatively prime to p , 9.5(ii) implies that $\text{cor}_H^G \text{res}_H^G$ is an isomorphism on $H^n(G, M)_{(p)}$. In particular, the restriction map induces a monomorphism $H^n(G, M)_{(p)} \hookrightarrow H^n(H, M)$. In order to describe the image of this monomorphism, we need some terminology.

If G is an arbitrary group, $H \subseteq G$ is a subgroup, and M is a G -module, we will say that an element $z \in H^*(H, M)$ is G -invariant if $\text{res}_{H \cap gHg^{-1}}^H z = \text{res}_{H \cap gHg^{-1}}^{gHg^{-1}} gz$ for all $g \in G$. In case $H \triangleleft G$, this just says that $z \in H^*(H, M)^{G/H}$. Note that if $z = \text{res}_H^G w$ for some $w \in H^*(G, M)$ then z is G -invariant; for $gz = \text{res}_{gHg^{-1}}^G w$ by exercise 1 of §9, hence $\text{res}_{H \cap gHg^{-1}}^{gHg^{-1}} gz = \text{res}_{H \cap gHg^{-1}}^G w = \text{res}_{H \cap gHg^{-1}}^H z$.

We can now state:

(10.3) Theorem. *Let G be a finite group and H a p -Sylow subgroup. For any G -module M and any $n > 0$, res_H^G maps $H^n(G, M)_{(p)}$ isomorphically onto the set of G -invariant elements of $H^n(H, M)$. In particular, if $H \triangleleft G$ then*

$$H^n(G, M)_{(p)} \approx H^n(H, M)^{G/H}.$$

PROOF. We have already shown that res_H^G maps $H^n(G, M)_{(p)}$ monomorphically into the G -invariants, so it remains to show that any invariant z is in the image. Consider the element $w = \text{cor}_H^G z \in H^n(G, M)$. Since $H^n(H, M)$ is annihilated by $|H|$, $w \in H^n(G, M)_{(p)}$. We now compute $\text{res}_H^G w$ by the double coset formula 9.5(iii); since z is invariant, we obtain

$$\begin{aligned} \text{res}_H^G w &= \sum_{g \in H \backslash G / H} \text{cor}_{H \cap gHg^{-1}}^H \text{res}_{H \cap gHg^{-1}}^{gHg^{-1}} gz \\ &= \sum_{g \in H \backslash G / H} \text{cor}_{H \cap gHg^{-1}}^H \text{res}_{H \cap gHg^{-1}}^H z \\ &= \sum_{g \in H \backslash G / H} (H: H \cap gHg^{-1}) z \\ &= (G: H) z. \end{aligned}$$

(The last equality is obtained by decomposing the set G/H into H -orbits and noting that $(H: H \cap gHg^{-1})$ is the cardinality of the H -orbit of gH .) Since $(G: H)$ is prime to p , it follows that $z = \text{res}_H^G w'$, where

$$w' = w/(G: H) \in H^n(G, M)_{(p)}. \quad \square$$

This proof, which should be thought of as an averaging argument, can be used in other situations where division by $(G: H)$ is possible. For example, one proves in the same way:

(10.4) Proposition. *Let G be arbitrary, M a G -module, and $H \subseteq G$ a subgroup of finite index. If $(G: H)$ is invertible in M then res_H^G maps $H^*(G, M)$ isomorphically onto the set of G -invariants in $H^*(H, M)$. In particular, if $H \triangleleft G$ then $H^*(G, M) \approx H^*(H, M)^{G/H}$.*

EXERCISES

1. Use 10.3 to compute $H^*(G, \mathbb{Z})$, where G is the symmetric group on three letters. [This will be easier after we have cup products available, cf. exercise 5 of §V.3, but it is doable now.]
2. Let $H \subseteq G$ be a subgroup of finite index. For any G -module M and any H - H double coset C , define a G -endomorphism $f(C)$ of $\mathbb{Z}G \otimes_{\mathbb{Z}H} M$ by $1 \otimes m \mapsto \sum_{g \in C/H} g \otimes g^{-1}m$. Using Shapiro's lemma, we can view the map $H^*(G, f(C))$ as an endomorphism $T(C)$ of $H^*(H, M)$.
 - (a) If $C = HgH$, show that $T(C)z = \text{cor}_{H \cap gHg^{-1}}^H \text{res}_{H \cap gHg^{-1}}^{HgHg^{-1}} z$. [Hint: It suffices to check this in dimension 0.]
 - (b) If $z \in H^*(H, M)$ is G -invariant, show that $T(C)z = a(C)z$, where $a(C) = |C/H| = (H: H \cap gHg^{-1})$.
 - (c) In the situation of 10.3 and 10.4, show that the image of res_H^G is the set of z such that $T(C)z = a(C)z$ for all double cosets C .

CHAPTER IV

Low-Dimensional Cohomology and Group Extensions

1 Introduction

An *extension* of a group G by a group N is a short exact sequence of groups

$$(*) \quad 1 \rightarrow N \rightarrow E \rightarrow G \rightarrow 1.$$

[Warning: Some people call this an extension of N by G .] A second extension $1 \rightarrow N \rightarrow E' \rightarrow G \rightarrow 1$ of G by N is said to be *equivalent* to $(*)$ if there is a map $E \rightarrow E'$ making the diagram

$$\begin{array}{ccccccc} & & & E & & & \\ & & \nearrow & \downarrow & \searrow & & \\ 1 & \longrightarrow & N & & G & \longrightarrow & 1 \\ & & \searrow & \downarrow & \nearrow & & \\ & & & E' & & & \end{array}$$

commute. Note that such a map is necessarily an isomorphism.

The main problem in the subject of group extensions is to classify the extensions of G by N up to equivalence. Roughly speaking, then, we are trying to understand all possible ways of building a group E with N as a normal subgroup and G as the quotient. This problem turns out to involve the cohomology functors $H^i(G, _)$ for $i = 1, 2, 3$.

For a while we will consider only the case where the kernel N is an abelian group A (written additively). A special feature of this case is that an extension

$$0 \rightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \rightarrow 1$$

gives rise to an *action* of G on A , making A a G -module. For E acts on A by conjugation since A is embedded as a normal subgroup of E ; and the conjugation action of A on itself is trivial, so there is an induced action of $E/A = G$

on A . Explicitly, given $g \in G$ we choose $\tilde{g} \in E$ such that $\pi(\tilde{g}) = g$, and the action of g on A is then characterized by

$$(1.1) \quad i(ga) = \tilde{g}i(a)\tilde{g}^{-1} \quad (a \in A).$$

It is often convenient to rewrite 1.1 as a *commutation rule*

$$(1.2) \quad \tilde{g}i(a) = i(ga)\tilde{g}.$$

In particular, this shows that $i(A)$ is central in E if and only if the G -action is trivial. In this case the extension is called a *central extension*.

In view of the G -module structure on A , we can refine the classification problem by fixing a G -module A and trying to classify the extensions of G by A which give rise to the given action of G on A . As we will see in §3, this problem has a very simple solution: The equivalence classes of such extensions are in 1-1 correspondence with the elements of $H^2(G, A)$.

We begin by reviewing the simplest class of extensions, namely, the *split extensions*.

2 Split Extensions

Fix a G -module A and let

$$(*) \quad 0 \rightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \rightarrow 1$$

be an extension which gives rise to the given action of G on A . We say that $(*)$ *splits* if there is a homomorphism $s: G \rightarrow E$ such that $\pi s = \text{id}_G$. The following characterization of split extensions is probably well-known to the reader, but we will write out the proof in detail since it motivates the proof of the main result of the next section.

(2.1) Proposition. *The following conditions on the extension $(*)$ are equivalent:*

- (i) $(*)$ *splits.*
- (ii) E *has a subgroup* $\tilde{G}$ *which is mapped by* π *isomorphically onto* G , *i.e., which satisfies* $E = i(A) \cdot \tilde{G}$ *and* $i(A) \cap \tilde{G} = \{1\}$.
- (iii) E *has a subgroup* $\tilde{G}$ *such that every element* $e \in E$ *is uniquely expressible in the form* $e = i(a)\tilde{g}$ $(a \in A, \tilde{g} \in \tilde{G})$.
- (iv) $(*)$ *is equivalent to the extension*

$$0 \rightarrow A \xrightarrow{i'} A \rtimes G \xrightarrow{\pi'} G \rightarrow 1,$$

where $A \rtimes G$ *is the semi-direct product of* G *and* A *relative to the given action, and* i' *and* π' *are the canonical inclusion and projection maps.*

(The definition of $A \rtimes G$ will be recalled in the course of the proof of 2.1.)

PROOF. Clearly (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii). To prove that these conditions imply (iv), let $s: G \rightarrow E$ be a splitting and note that we have a set-theoretic bijection $A \times G \approx E$, given by $(a, g) \mapsto i(a)s(g)$. There is therefore a unique group law on the set $A \times G$ such that this bijection is an isomorphism. To calculate this group law, we must express a typical product $[i(a)s(g)] \cdot [i(b)s(h)]$ in E in the form $i(-)s(-)$. Using the commutation rule 1.2, we find

$$\begin{aligned} i(a)s(g)i(b)s(h) &= i(a)i(gb)s(g)s(h) \\ &= i(a + gb)s(gh). \end{aligned}$$

Thus the group law on $A \times G$ is given by

$$(a, g) \cdot (b, h) = (a + gb, gh).$$

The set $A \times G$ with this group law is, by definition, the semi-direct product $A \rtimes G$, and (iv) follows at once. Finally, it is trivial that (iv) $\Rightarrow$ (i). $\square$

Proposition 2.1 says that there is only one split extension of G by A (up to equivalence) associated to the given action of G on A . Nevertheless, there is an interesting “classification” problem involving split extensions: Given that an extension $(*)$ splits, classify all possible splittings.

In case G acts trivially on A , for example, so that the group E is isomorphic to the direct product $A \times G$, then the splittings are obviously in 1-1 correspondence with homomorphisms $G \rightarrow A$. In the general case, I claim that splittings correspond to *derivations* (also called *crossed homomorphisms*). These are functions $d: G \rightarrow A$ satisfying

$$(2.2) \quad d(gh) = dg + g \cdot dh$$

for all $g, h \in G$. To see this we may assume that $(*)$ is the canonical split extension

$$0 \rightarrow A \rightarrow A \rtimes G \rightarrow G \rightarrow 1.$$

A function $s: G \rightarrow A \rtimes G$ with $\pi s = \text{id}$ then has the form $s(g) = (dg, g)$, where d is a function $G \rightarrow A$. We have

$$s(g)s(h) = (dg + g \cdot dh, gh),$$

so s will be a homomorphism if and only if d is a derivation. This proves the claim.

Remark. The term “derivation” seems more reasonable if we think of G as acting on A on the right by the trivial action, in addition to the given left action of G on A . The equation 2.2 then takes the familiar form

$$d(gh) = dg \cdot h + g \cdot dh,$$

as in the product rule for derivatives.

Two splittings s_1, s_2 will be said to be *A-conjugate* if there is an element $a \in A$ such that $s_1(g) = i(a)s_2(g)i(a)^{-1}$ for all $g \in G$. Since

$$(a, 1)(b, g)(a, 1)^{-1} = (a + b - ga, g)$$

in $A \rtimes G$, this conjugacy relation becomes

$$d_1 g = a + d_2 g - ga$$

in terms of the derivations d_1, d_2 corresponding to s_1, s_2 . Thus d_1 and d_2 correspond to *A-conjugate* splittings if and only if their difference $d_2 - d_1$ is a function $G \rightarrow A$ of the form $g \mapsto ga - a$ for some fixed $a \in A$. Such a function is said to be a *principal derivation*.

We can summarize the preceding paragraph by saying that the *A-conjugacy* classes of splittings of a split extension of G by A correspond to the elements of the quotient group $\text{Der}(G, A)/P(G, A)$, where $\text{Der}(G, A)$ is the abelian group of derivations $G \rightarrow A$ and $P(G, A)$ is the group of principal derivations. On the other hand, a glance at the standard cochain complex $C^*(G, A)$ shows that $\text{Der}(G, A)$ is the group of 1-cocycles and $P(G, A)$ is the group of 1-coboundaries (cf. exercise 2 of §III.1). We have therefore established:

(2.3) Proposition. *For any G -module A , the A -conjugacy classes of splittings of the split extension*

$$0 \rightarrow A \rightarrow A \rtimes G \rightarrow G \rightarrow 1$$

are in 1-1 correspondence with the elements of $H^1(G, A)$.

EXERCISES

The purpose of these exercises is to develop the theory of derivations and to give algebraic proofs of some results that were proved topologically in Chapters I and II.

1. If d is a derivation, show that $d(1) = 0$.
2. Let I be the augmentation ideal of $\mathbb{Z}G$ and let $D: G \rightarrow I$ be the derivation defined by $Dg = g - 1$ for $g \in G$. [Regarded as a derivation $G \rightarrow \mathbb{Z}G$, this is simply the principal derivation corresponding to $1 \in \mathbb{Z}G$.] Show that D is the *universal derivation* on G , in the following sense. Given any G -module A and any derivation $d: G \rightarrow A$, there is a unique G -module map $f: I \rightarrow A$ such that $d = fD$:

$$\begin{array}{ccc} G & \xrightarrow{D} & I \\ \downarrow d & \searrow f & \\ A & & \end{array}$$